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Faculty of Computing

AI-Based Heart Disease Prediction System Using Artificial Neural Networks

Final Report

Essentials of Artificial Intelligence (CS22023)

Group No. 42

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Source code: github.com/nethmishehara2004-svg/HeartDiseasePrediction

Live prototype: heart-disease-prediction-abfmrgf6nvvbwlsigeurtk.streamlit.app

Abstract

Heart disease is the cause of death of millions of people around the globe, and a prediction of the risk factors involves taking into account multiple clinical parameters of patients. In this project, a prototype of an AI system for preliminary diagnosis of heart disease is introduced. For that purpose, a supervised Artificial Neural Network (ANN) model was trained using a database of 1,024 patient cases with 13 clinical features. The inputs were standardized with the StandardScaler and the ANN (with three hidden layers consisting of 64, 32 and 16 neurons each with Batch Normalization and Dropout and a sigmoid activation function at the output layer) was optimized with Adam optimizer, early stopping, and learning rate reduction. On the test set (205 cases), the saved model produced 83.4% accuracy, 81.3% precision, 83.0% recall, 82.1% F1-score and 0.921 ROC-AUC score. The model is used within a Streamlit web application (and a Tkinter desktop application version) where the user inputs their health data and obtains a prediction with a probability. The prototype is an educational and not medical decision-support tool. The project report includes discussion about its limitations that include a synthetic nature of the dataset, relatively small size of the test dataset and an issue with the encoding of the inputs revealed in the final testing.

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Chapter 1: Introduction

1.1 Introduction

Cardiovascular disease is responsible for many deaths around the world [1], and early detection of risk enables early intervention by healthcare professionals. There have been extensive studies done using machine learning in this area, comparing classical techniques like random forests and deep neural networks on public datasets containing heart diseases [2], [3], [4], [5], [6], [7], as well as the influence of feature selection on accuracy [8], [9]. Inspired by the above studies, we developed an end-to-end prototype that involves all the steps of the AI process: data preprocessing, designing the model, training, evaluation, and deploying the result via a user interface.

This study fulfills the requirement set out by the assignment, which includes several AI-related concepts: supervised learning, classification using neural networks, data preprocessing including feature scaling, regularization, and model evaluation based on decision-making using thresholds.

1.2 Problem Background

Whether a person has heart disease depends on many interacting factors, including age, sex, chest-pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG result, maximum heart rate, exercise-induced angina, ST depression, the slope of the ST segment, the number of major vessels coloured by fluoroscopy and a thalassemia test result. Clinicians assess these together, which takes expertise and time. People without easy access to that expertise have little way to get even a rough first indication of their risk.

Currently, risk is assessed by clinicians reviewing test results, sometimes supported by scoring charts. The limitations are that manual evaluation is time-consuming, patterns across many parameters are hard to see, and a preliminary screen is not always r The determination of whether an individual suffers from heart disease is done considering numerous variables such as age, sex, type of chest pain, blood pressure at rest, cholesterol, fasting blood sugar, ECG test result at rest, maximum heart rate, exercise-induced angina, depression of ST, the slope of ST segment, number of major vessels colored by fluoroscopy and thalassemia test result. The analysis of these factors requires time and expertise. People who do not have access to these experts have no choice but to wait to find out their chances of suffering from heart diseases.

Risk assessment is currently done by clinical experts analyzing the test results, which might be assisted by scoring tables. Some of the problems associated with this approach include time consumption, inability to identify any patterns among various parameters and lack of any preliminary screening process. AI could help identify the relationship between the parameters using historical patient data and generate an initial estimate for a patient.eadily available. AI can

learn the relationships between these parameters from historical patient records and return a fast, consistent preliminary estimate for a new patient.

1.3 Problem Statement

Manual analysis of a number of characteristics of patients takes too much time and does not allow discovering patterns within the set of health parameters. In the course of the current project, an artificial intelligence system that will be based on the Artificial Neural Network and designed to analyze 13 clinical parameters and forecast the probability of developing heart disease has been developed.

1.4 Objectives

1. Design an ANN model to determine the existence of heart disease based on 13 clinical variables.
2. Sample a dataset containing more than 1,000 entries and carry out the pre-processing of the sampled data by checking for any missing, duplicate, and imbalanced data.
3. Perform data pre-processing of the input data through standardizing all the data in order to be compatible with the neural network due to their different scales.
4. Accurately measure the accuracy, precision, recall, F1-score, AUC-ROC curve, and confusion matrix as performance measures for training and testing the ANN.
5. Avoid overfitting the ANN using Batch Normalization, Dropout, early stopping, and learning rate reduction techniques.
6. Design an interface that will allow users to input health information and receive the output and probability of the prediction result.
7. Create awareness regarding heart disease risk factors without giving any medical advice.

1.5 Target Users

Individuals in the general population interested in knowing their probability; individuals with high blood pressure or cholesterol levels; students in healthcare or computing; or any researcher/professional looking for an additional demonstration aid.

Chapter 2: AI Techniques

2.1 Selected AI Techniques

Learning type and Classification problem. This being a classification problem involving both inputs and outputs (labels), this falls under the category of supervised learning / binary classification problem. Artificial Neural Network (ANN). It consists of feed-forward fully-connected neural network that learns a non-linear function to map the 13 inputs to the probability of having heart disease. The output neuron uses sigmoid activation to produce a value ranging from 0 to 1 which gets labelled depending on a threshold value of 0.5.

Data pre-processing and normalization. StandardScaler (scikit-learn) scales all the variables to have a mean of 0 and unit standard deviation. Regularization and Training Parameters. Batch Normalization, Dropout, early stopping and learning rate decay on plateau. Evaluation Metrics. Precision, Recall, F1 score, accuracy, AUC and Confusion Matrix.

2.2 Justification of AI Techniques

The risk of heart disease depends on the interaction of the factors (the interplay between age and exercise-induced angina and ST depression, for instance) and not on the individual factor. This interaction can be represented by a neural network that uses non-linear (ReLU) hidden layers without writing special rules. The model can be retrained using new data. A probabilistic result is preferable for the risk calculation as the interface allows showing the confidence of the predictions.

Input normalisation is important in our case because the ranges of values of the variables differ dramatically (cholesterol may be up to 564 while oldpeak – only up to 6.2). Otherwise, large-valued factors will affect the learning process much more than small-valued ones. The dropout technique disables some neurons during training randomly while batch normalization provides stability to the inputs of the layers. Both of them helped to reduce the gap between the performances on the training and validation data sets found at the beginning of work.

Accuracy as the sole metric of quality may be misleading when it comes to medical applications as there is a bigger price for missing a case. That is why recall, precision, F1 and AUC metrics are provided along with the accuracy.

Chapter 3: System Design

3.1 Workflow

1. **The Dataset** (heart_disease.csv, 1,024 records) is loaded and analyzed.
2. **Data preprocessing**: The helper variable 'num' is removed, features and target are segregated and then the dataset is split into training, validation and testing datasets.
3. **Standardization**: The Standard Scaler is fit only on the training set and is saved as scaler.pkl.
4. **Model training**: The ANN model is trained and saved as heart_model.keras.
5. **Model Evaluation**: The trained model is evaluated using the unseen test set.
6. **Model deployment**: In this phase, the deployed system loads the pre-saved model and standardizer. User inputs are transformed to numerical codes and then fed into the ANN via the pre-loaded scaler and the result is returned as a probability.

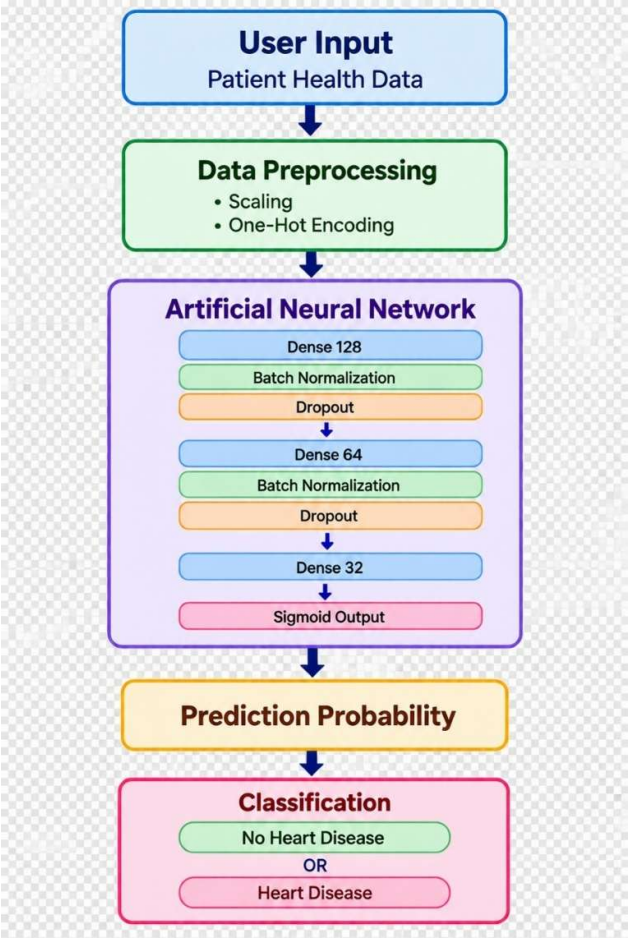


Figure 1: System architecture of the heart disease prediction system

3.2 Dataset

The 13 attributes in the clinical data inputs correspond to the attribute list of the commonly used UCI Heart Disease (Cleveland) database [10], [11]. This dataset contains 1,024 entries and 15 variables – 13 attributes in the clinical data inputs, original diagnosis num (0-4), and the target attribute target_binary. Attribute num is stripped out before training, while target_binary is the target for prediction.

Table 1: Input features

Feature	Description	Values in dataset
age	Age in years	18 – 78.8
sex	Sex	0 = female, 1 = male
cp	Chest pain type	1 = typical angina, 2 = atypical angina, 3 = non-anginal pain, 4 = asymptomatic
trestbps	Resting blood pressure (mm Hg)	84.9 – 200
chol	Serum cholesterol (mg/dl)	100 – 564
fbs	Fasting blood sugar > 120 mg/dl	0 = no, 1 = yes
restecg	Resting ECG result	0 = normal, 1 = ST-T abnormality, 2 = LV hypertrophy
thalach	Maximum heart rate achieved	71 – 227.2
exang	Exercise-induced angina	0 = no, 1 = yes
oldpeak	ST depression induced by exercise	0 – 6.2
slope	Slope of peak exercise ST segment	1 = upsloping, 2 = flat, 3 = downsloping
ca	Major vessels coloured by fluoroscopy	0 – 3
thal	Thalassemia test result	3 = normal, 6 = fixed defect, 7 = reversible defect

Table 2: Target distribution

Target	Meaning	Records	Share
0	No Heart Disease	554	54.1%
1	Heart Disease	470	45.9%

The classes are reasonably balanced, so no resampling was needed.

3.3 Data Split

The data set was partitioned using a constant random seed value (42) and stratifying based on the label such that the proportion of classes is maintained in each partition: test (20%) – 205 rows, followed by validation (20%) – 164 rows, leaving 655 rows for training.

3.4 Neural Network Structure

Table 3: ANN architecture

Layer	Output shape	Parameters
Input	13	–
Dense (ReLU)	64	896
Batch Normalization	64	256
Dropout (0.20)	64	0
Dense (ReLU)	32	2,080
Batch Normalization	32	128
Dropout (0.15)	32	0
Dense (ReLU)	16	528
Dense (sigmoid)	1	17
Total		**3,905** (3,713 trainable, 192 non-trainable)

Chapter 4: Implementation

4.1 Tools and Libraries

Python; TensorFlow/Keras for the artificial neural network; Scikit-learn for the splitting, scaling and metrics; Pandas and NumPy for the data manipulation; Matplotlib for plotting; Joblib for the save scaler; Streamlit for the Web interface; Tkinter for the Desktop interface; GitHub for the version control.

4.2 Training Configuration

Table 4: Training configuration

Setting	Value
Optimiser	Adam [12], learning rate 0.001
Loss	Binary cross-entropy
Batch size	32
Maximum epochs	150
Early stopping	Monitor validation AUC, patience 15, restore best weights
Learning-rate reduction	Monitor validation loss, factor 0.5, patience 5, minimum 0.00001
Metrics tracked	Accuracy, AUC, precision, recall
Decision threshold	0.5
Random seeds	NumPy and TensorFlow seed 42

4.3 Changes Since the Progress Review

The progress report presented a neural network of neurons in the range 32-16, batch size of 16, epochs of 100, and one hot encoding for the categorical attributes. In the actual implementation, the network is the 64-32-16 mentioned above, a batch size of 32, 150 maximum epochs with early stopping, and the 13 numeric codes are standardized directly without one-hot encoding. This leaves the number of features in the input as 13, consistent with the interface. The proposal also suggested smoking status, diabetes status, and BMI as inputs; these have been omitted as the chosen data set doesn't have them.

4.4 User Interfaces

The two interfaces are based on the same saved model and scaler. The first interface is Streamlit web app (app.py) and the second one is Tkinter desktop application (gui.py). Both interfaces provide 13 input fields where dropdown menus are used to select categorical variables and numeric inputs to measure other variables and show output “Heart disease detected” or “No heart disease detected” with a level of confidence. The confidence provided is the probability of the predicted class from the model.

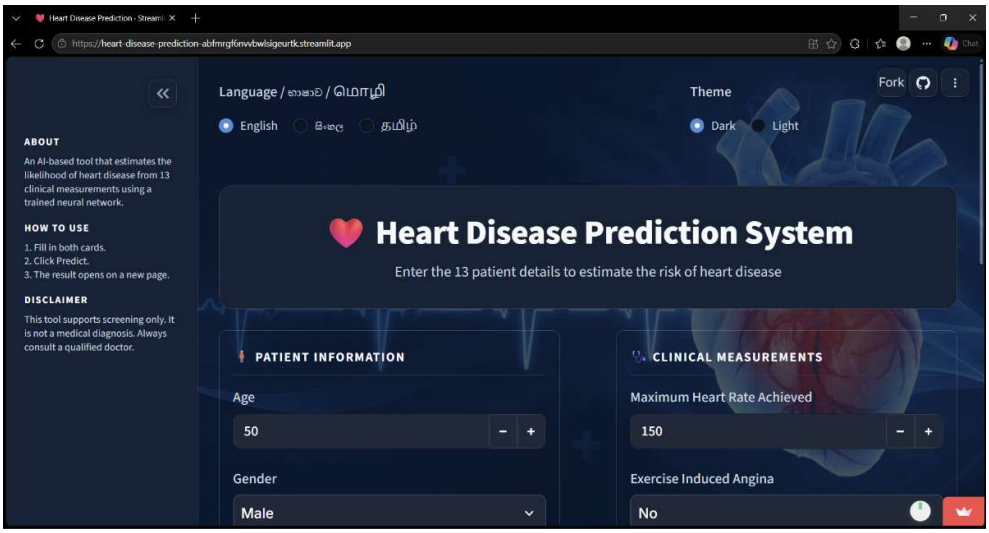


Figure 2: Streamlit web interface

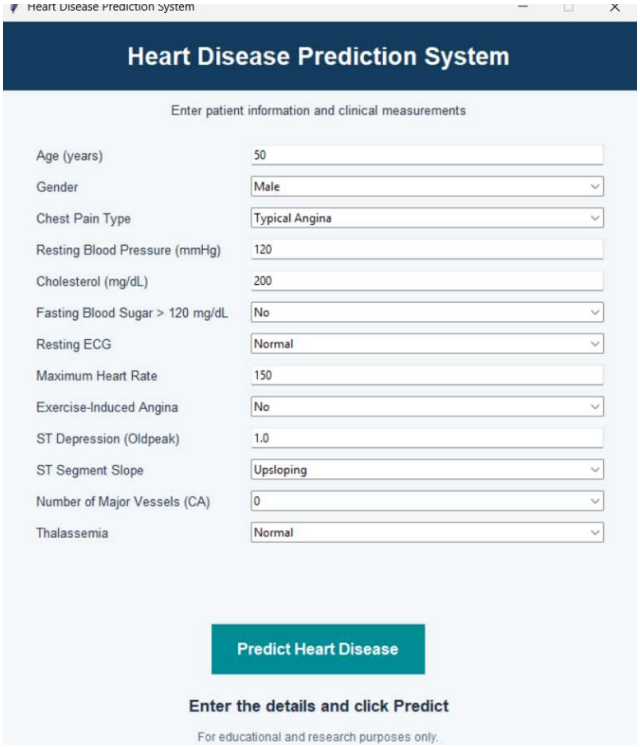


Figure 3: Tkinter desktop interface

Chapter 5: Testing and Results

5.1 Training Behaviour

However, in the training process, there was early stoppage at epoch 53 because of overfitting. This resulted in an accuracy of 93.3% on training data while the validation accuracy was 89.6%. The value of validation AUC was 0.947 in this case. While some amount of overfitting remains despite regularisation, the difference between validation accuracy and training accuracy is far lesser as compared to the group's earlier efforts that did not use Dropout and Batch Normalization.

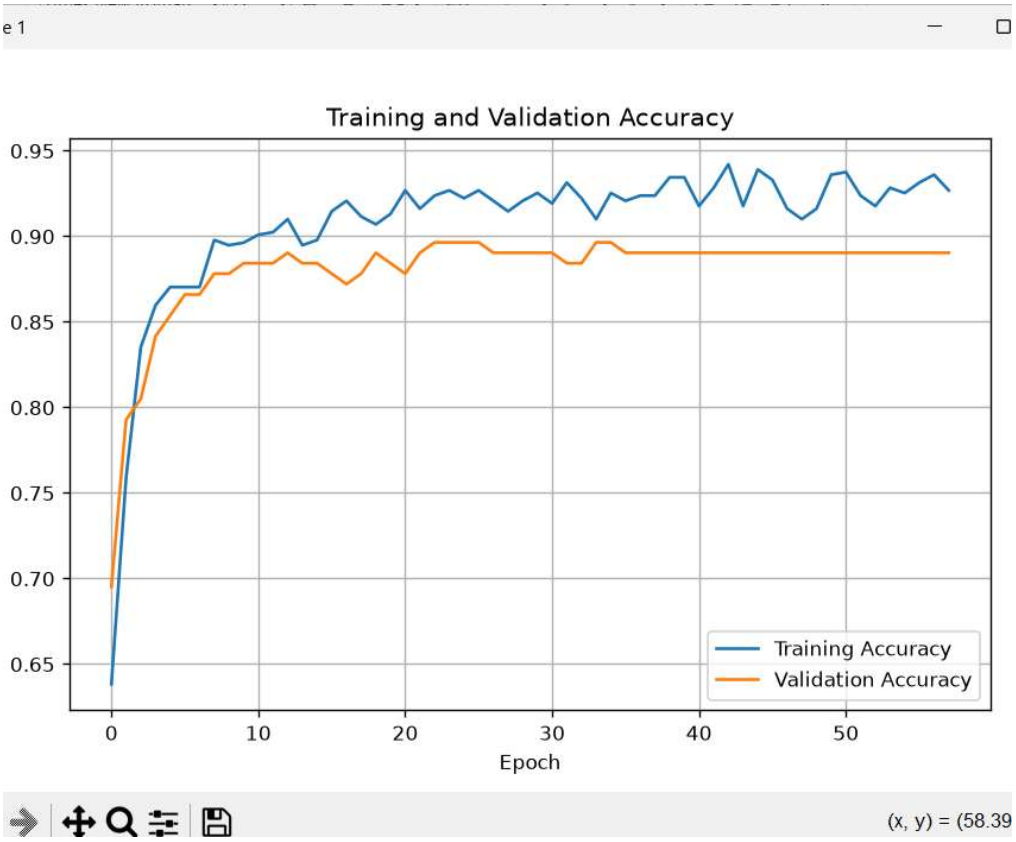


Figure 4: Training and validation accuracy

5.2 Test-Set Performance

The saved model heart_model.keras was then tested on 205 test data points using a threshold value of 0.5.

Table 5: Test-set metrics

Metric	Value
Accuracy	83.4%
Precision (Heart Disease)	81.3%
Recall (Heart Disease)	83.0%
F1-score	82.1%
ROC-AUC	0.921

Table 6: Confusion matrix

	Predicted: No Heart Disease	Predicted: Heart Disease
Actual: No Heart Disease	93	18
Actual: Heart Disease	16	78

However, in the process of analysis, the algorithm was able to identify 78 out of 94 patients that suffered from heart diseases (recall 83.0%) and 93 out of 111 patients who didn't have heart diseases. 16 patients were classified incorrectly and 18 false alarms occurred. Another analysis that was conducted by running again the training script in another software produced almost the same but not identical results (accuracy of 83.9%, AUC 0.920). It is due to the fact that neural network training is not entirely deterministic in various versions of libraries. The confusion matrix (95/16/19/75) used for the progress review has been created before.

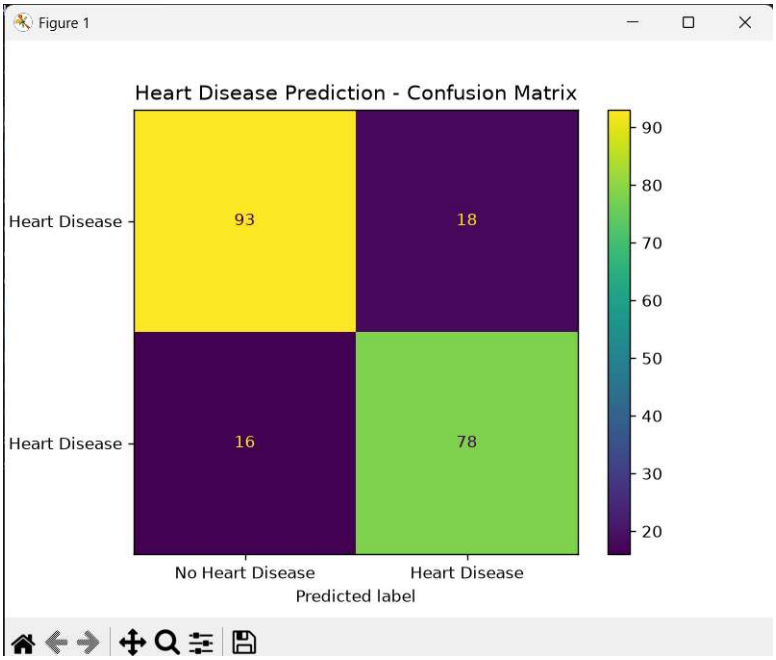


Figure 5: Confusion matrix of the model on the test set

5.3 Sample Predictions

Eight instances of the test data set have been fed into the model (Probability of heart disease > 0.5). Ages, blood pressure, cholesterol, and heart rate are rounded off. “Pred.” and “Actual” indicate whether there was a prediction of heart disease or if it had occurred.

Table 7: Sample test-set predictions

#	Age	Sex	cp	BP	Chol	Max HR	Exang	Oldpeak	Slope	ca	thal	Prob.	Pred.	Actual
1	34	M	1	118	182	174	0	0.0	1	0	3	0.001	No	No
2	61	M	4	121	246	101	1	0.0	2	0	3	0.757	Yes	Yes
3	52	F	3	136	196	169	0	0.1	2	0	3	0.038	No	No
4	62	F	4	160	314	144	0	2.1	1	0	7	0.675	Yes	Yes
5	70	M	3	133	344	122	0	0.3	1	1	3	0.077	No	No
6	60	M	2	153	260	158	1	1.1	3	2	7	0.977	Yes	Yes
7	51	M	3	94	227	154	1	0.0	1	1	7	0.611	Yes	No
8	73	F	3	135	269	128	1	2.2	2	0	3	0.745	Yes	No

All the eight predictions coincide with the actual result, except for Samples 7 and 8, and the outputs make clinical sense: low probability of younger people who do not have any angina or ST depression, while a high probability occurs where there are all three factors. Samples 7 and 8 represent false positives; this is why the output cannot be considered a diagnosis.

5.4 Threshold Analysis

The application predicts heart disease when the model's output probability is 0.5 or higher. This cutoff is called the decision threshold. Because missing a patient with heart disease is more serious than a false alarm, the saved model was evaluated on the same 205 unseen test patients at thresholds from 0.2 to 0.8 to see how the results change. The results are shown in Table 8 and Figure 6.

Table 8: Model performance at different decision thresholds

Threshold	Missed disease(FN)	False alarms(FP)	Accuracy	Precision	Recall	F1
0.2	6	33	0.810	0.727	0.936	0.819
0.3	13	26	0.810	0.757	0.862	0.806
0.4	16	23	0.810	0.772	0.830	0.800
0.5 (Default)	16	18	0.834	0.812	0.830	0.821
0.6	17	13	0.854	0.856	0.819	0.837
0.7	24	10	0.834	0.875	0.745	0.805
0.8	31	4	0.829	0.940	0.670	0.783

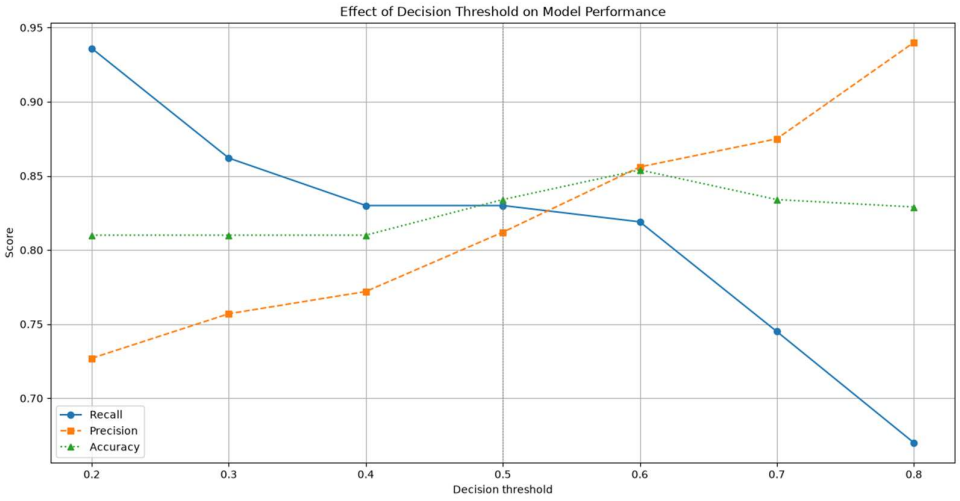


Figure 6: Threshold chart

5.5 Test Cases

The outcomes of each test case were predetermined based on the known risk factors of heart disease, such as age, blood pressure, cholesterol level, type of chest pain, ST depression, and number of affected vessels. In patients with combined risk factors, the expected outcome was "uncertain".

Table 9: Test Cases

Test ID	Description	Inputs	Expected	Actual	Confidence	Pass/ Fail
TC - 01	App loads and shows default values (Baseline)	Open the app. No input. Default shown: Age 50, Male, Typical Angina, BP 120, Chol 200, FBS No, ECG Normal, Max HR 150, Ex Angina No, Old peak 1.00, Upsloping, Vessels 0, Thal Normal	Page loads with no error. Both input cards and the Predict button show, with default values filled in.	Page loaded. Input cards and Predict button showed with default values.	N/A	Pass
TC - 02	Baseline prediction	Default values	A prediction result. Probability, Confidence, Risk level show on the result page without errors.	Results are shown without any errors. Probability 17.0%, Low risk.	83.0%	Pass

Test ID	Description	Inputs	Expected	Actual	Confidence	Pass/ Fail
TC - 03	Low-risk patient	Age 35, Female, Non-anginal, BP 110, Chol 170, FBS No, ECG Normal, Max HR 175, Angina No, ST 0.0, Upsloping, Vessels 0, Thal Normal	No Disease / Low risk	No Disease Detected. Low risk.	93.8%	Pass
TC - 04	High-risk patient	Age 65, Male, Atypical Angina, BP 160, Chol 290, FBS Yes, ECG ST-T abn., Max HR 110, Angina Yes, ST 3.0, Flat, Vessels 3, Thal Reversible defect	Disease / High risk	Heart disease detected / High risk. Probability 52.7% (just above the 0.5 threshold).	52.7%	Pass
TC - 05	Higher age	Default values with age 70	Probability higher than TC - 02	No heart disease detected. Probability 19.0% (TC-02 = 17.0%).	81.0%	Pass
TC - 06	Asymptomatic chest pain	Baseline, but chest pain asymptomatic	No disease. Probability higher than TC - 02	No heart disease detected. Probability 14.3% (TC-02 = 17.0%), so the probability was lower, not higher.	85.7%	Fail
TC - 07	More major vessels	Default values with vessels 3	No heart disease. Probability higher than TC-02	No heart disease detected. Probability 30.9%.	69.1%	Pass
TC - 08	Exercise angina	Baseline values with Angina YES	No disease. Probability higher than TC-02	No heart disease detected. Probability 18.2%.	81.8%	Pass
TC - 09	High ST depression	Baseline values with ST 3.0	No disease. Probability higher	No heart disease detected.	78.1%	Pass

Test ID	Description	Inputs	Expected	Actual	Confidence	Pass/ Fail
			than TC-02	Probability 21.9%.		
TC - 10	High max heart rate	Baseline with Max HR 190	No heart disease detected. Probability lower than or equal to TC-02	No heart disease detected. Probability 12.6%.	87.4%	Pass
TC - 11	Gender change	Baseline values with female	No disease (or disease if the probability reaches 50%). A result shows without errors.	No heart disease detected. Probability 16.2%.	83.8%	Pass
TC - 12	Age at minimum	Default values with age at the field's minimum value (Age 1)	Accepted. A prediction (disease or no disease) shows with no error.	Accepted. Result shown (No heart disease. Prob = 6.3%).	93.7%	Pass
TC - 13	Repeatability	Age 62, Male, Non-anginal, BP 120, Chol 200, FBS Yes, ECG Normal, Max HR 150, Angina No, ST 1.00, Upsloping, Vessels 1, Thal Fixed defect. Click Predict twice.	The same result both times.	No heart disease detected. Probability is 20.9% both times (Moderate risk).	79.1%	Pass
TC - 14	Dark mode and Light mode	Age 69, Female, Asymptomatic, BP 190, Chol 280, FBS Yes, ECG ST-T abn., Max HR 90, Angina Yes, ST 2.50, Flat, Vessels 2, Thal Reversible defect. Switch Theme to Light at the top of the page, then click Predict.	All text stays readable in light mode. A prediction result shows without errors.	Texts are readable. Heart disease detected / High risk. Probability 50.5%.	50.5%	Pass
TC - 15	Return and re-enter	From the result page, go back and enter the TC-03 values.	The form works again. The result matches TC-03.	The form worked again. The values were entered as in TC-03 but with FBS =	94.2%	Fail

Test ID	Description	Inputs	Expected	Actual	Confidence	Pass/ Fail
				Yes. Probability is 5.8% (TC-03 = 6.2%), so the result did not match TC-03.		
TC - 16	Sidebar content	Open the sidebar.	About, How to use and Disclaimer sections show correctly in the sidebar.	Showed the content of the sidebar (About, How to use and Disclaimer).	N/A	Pass
TC - 17	PDF button visible and works	Default values. Click Predict.	A download PDF report button shows on the result page.	A PDF file downloaded without any errors and opened in another tab.	N/A	Pass
TC - 18	PDF matches the result page	Default values. Open the downloaded PDF.	The PDF shows the same class (No Disease), probability (17.0%), confidence and risk level as the result page.	The PDF showed the same results as the system.	83.0%	Pass
TC - 19	Language selector shows all three languages	Open the app. No input.	English, සිංහල and *தமிழ்* options show, with English selected by default.	English, සිංහල and *தமிழ்* options show at the top of the page, with English selected by default.	N/A	Pass
TC - 20	Switch to Sinhala (input page)	Default values. Select සිංහල.	The title, all 13 field labels, dropdown options and Predict button appear in Sinhala. No broken characters or empty boxes.	All 13 field labels, dropdown options and both buttons show in Sinhala with no broken	N/A	Pass

Test ID	Description	Inputs	Expected	Actual	Confidence	Pass/ Fail
				characters (Figures 44 and 47). The page title also shows in Sinhala (Figure 47).		
TC - 21	Switch to Tamil (input page)	Default values. Select தமிழ்.	The title, all 13 field labels, dropdown options and Predict button appear in Tamil. No broken characters or empty boxes.	All 13 field labels, dropdown options and both buttons show in Tamil with no broken characters (Figures 45 and 48). The page title also shows in Tamil (Figure 48).	N/A	Pass
TC - 22	Same default values in all languages	Default values. View the input page in English, then in Sinhala, then in Tamil.	The same default values (Age 50, Male, Typical Angina, BP 120, Chol 200, FBS No, ECG Normal, Max HR 150, Angina No, ST 1.00, Upsloping, Vessels 0, Thal Normal) show in all three languages.	The same default values showed in English, Sinhala and Tamil (Figures 46 to 48).	N/A	Pass
TC - 23	Sinhala result page	Default values. Language Sinhala. Click Predict.	The result title, “Disease probability”, “Confidence”, risk level and patient detail labels show in Sinhala. The numbers match TC-02 (17.0%,	The result title, probability, confidence, risk level and patient details show in Sinhala. Probability 17.0%, confidence 83.0%, risk	83.0%	Pass

Test ID	Description	Inputs	Expected	Actual	Confidence	Pass/ Fail
			83.0%, Low).	level Low (same as TC-02).		
TC - 24	Tamil result page	Default values. Language Tamil. Click Predict.	The result title, “Disease probability”, “Confidence”, risk level and patient detail labels show in Tamil. The numbers match TC-02 (17.0%, 83.0%, Low).	The result title, probability, confidence, risk level and patient details show in Tamil. Probability 17.0%, confidence 83.0%, risk level Low (same as TC-02).	83.0%	Pass
TC - 25	Dropdown options translated	Default values. Open the Chest pain type dropdown in English, then in Sinhala, then in Tamil.	All four chest pain options show translated in each language, in the same order (Typical, Atypical, Non-anginal, Asymptomatic).	All four options showed translated in each language, in the same order. Typical Angina stayed selected (Figures 51 to 53).	N/A	Pass
TC - 26	Change language on the result page	Default values. Click Predict in English, then switch the language to Sinhala on the result page.	No error. The result stays or resets cleanly, and all labels change to the selected language.	The result stayed on the page (17.0%, 83.0%) and all labels changed to Sinhala (Figures 54 and 55).	83.0%	Pass
TC - 27	Out-of-range warning in Sinhala and Tamil	Default values with vessels 3 (same as TC-07). Predict in Sinhala, then in Tamil.	The “Values outside the typical range” warning for 3 major vessels shows translated. The result is the same as in English (TC-	The warning (3 major vessels affected) showed in Sinhala and Tamil. Probability 30.9%,	69.1%	Pass

Test ID	Description	Inputs	Expected	Actual	Confidence	Pass/ Fail
			07: 30.9%, Moderate).	confidence 69.1%, risk level Moderate in both languages (same as TC-07).		

Note: The prediction was made for Age 1 by the application. This age range [min, max] is not in the training data set, so predictions for these ages are not reliable.

27 GUI tests were run: 25 passed and 2 failed (TC-06 and TC-15). The predictions were consistent with low-risk and high-risk inputs, repeated inputs produced the same output, and the Sinhala and Tamil interfaces gave the same prediction as the English interface (17.0% disease probability and 83.0% confidence for the default values in all three languages). The confidence ranged from 50.5% to 94.2%. The lowest values were for the two high-risk cases (TC-04 and TC-14), which are close to the 0.5 threshold, and TC-07 (3 major vessels) gave 69.1%. TC-06 failed because asymptomatic chest pain gave a lower probability (14.3%) than the baseline (17.0%), which agrees with the input-encoding mismatch described in Section 6.2. TC-15 failed because the re-entered values (with FBS = Yes) gave 5.8% instead of the 6.2% obtained in TC-03. Age 1 was accepted by the application even though it is outside the training range, but a prediction was still made.

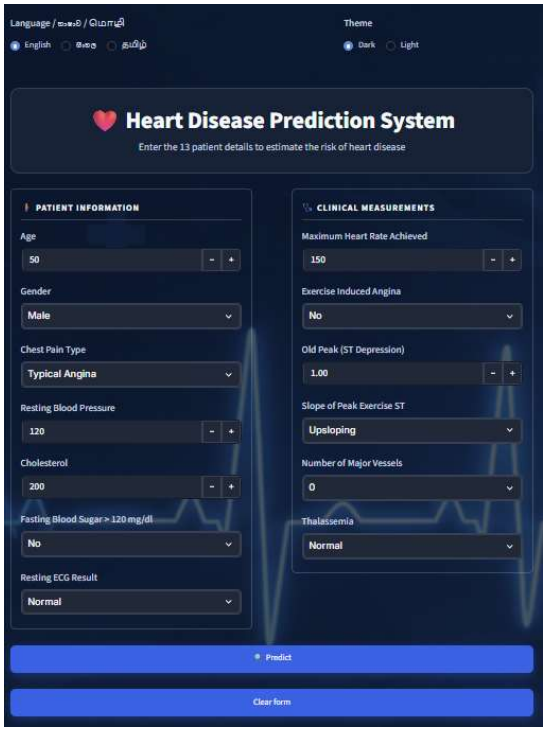


Figure 7: TC-01 – App loads with default values



Figure 8: TC-02 – Baseline prediction result



Figure 9: TC-03 – Low-risk patient result

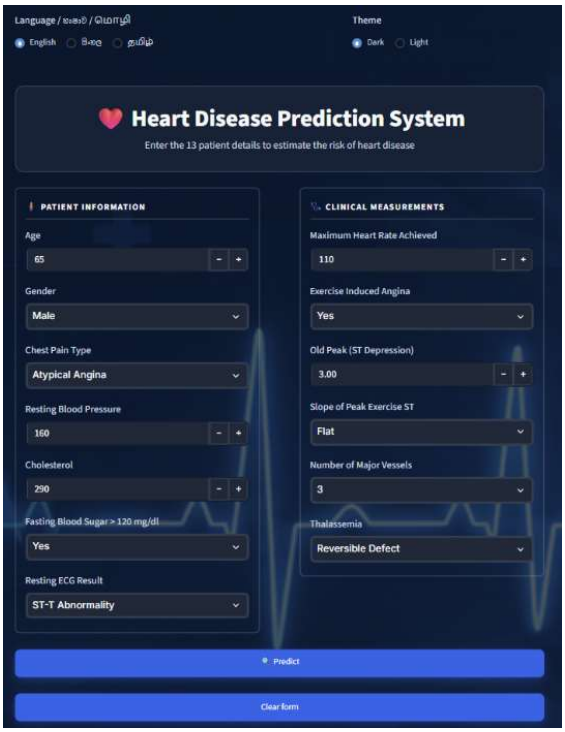


Figure 10: TC-04 – High-risk patient inputs



Figure 11: TC-04 – High-risk patient result



Figure 12: TC-04 – Patient details and download report button

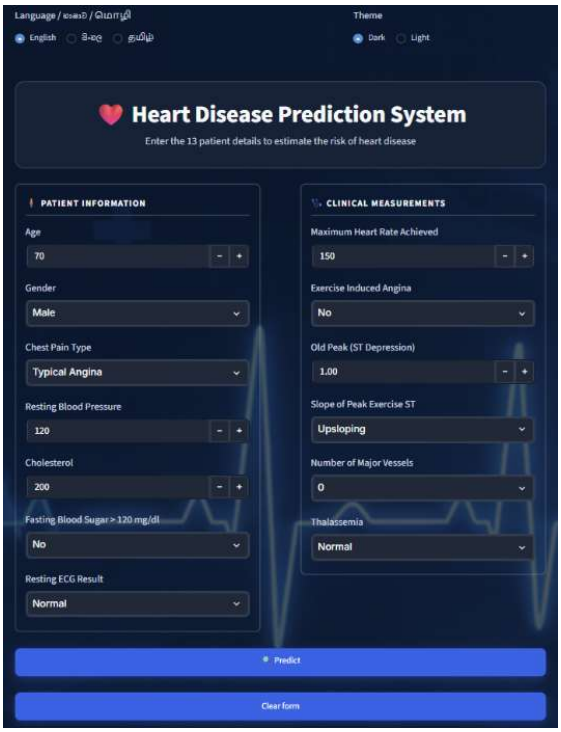


Figure 13: TC-05 – Higher age (70) inputs



Figure 14: TC-05 – Higher age (70) result

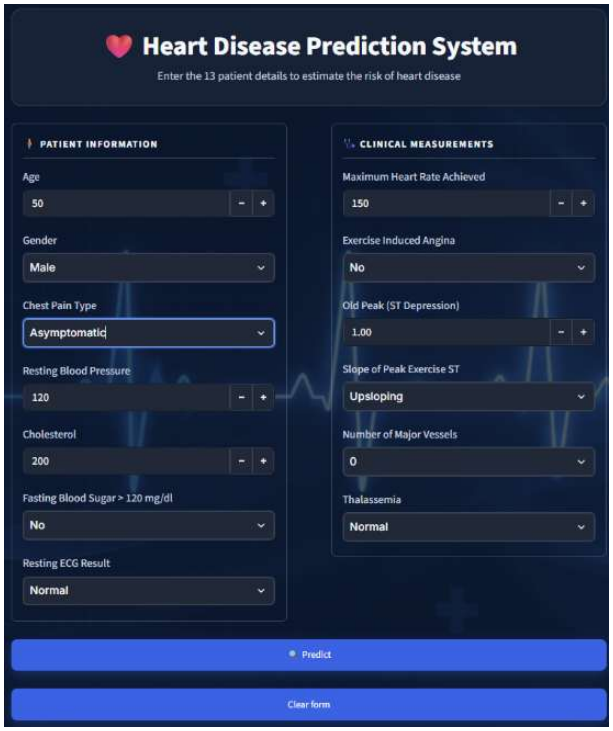


Figure 15: TC-06 – Asymptomatic chest pain inputs



Figure 16: TC-06 – Asymptomatic chest pain result

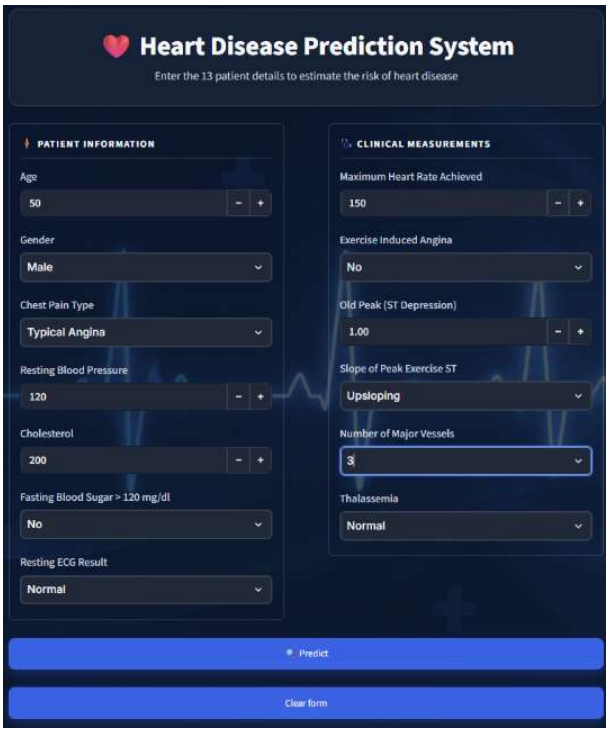


Figure 17: TC-07 – Major vessels = 3 inputs



Figure 18: TC-07 – Major vessels = 3 result

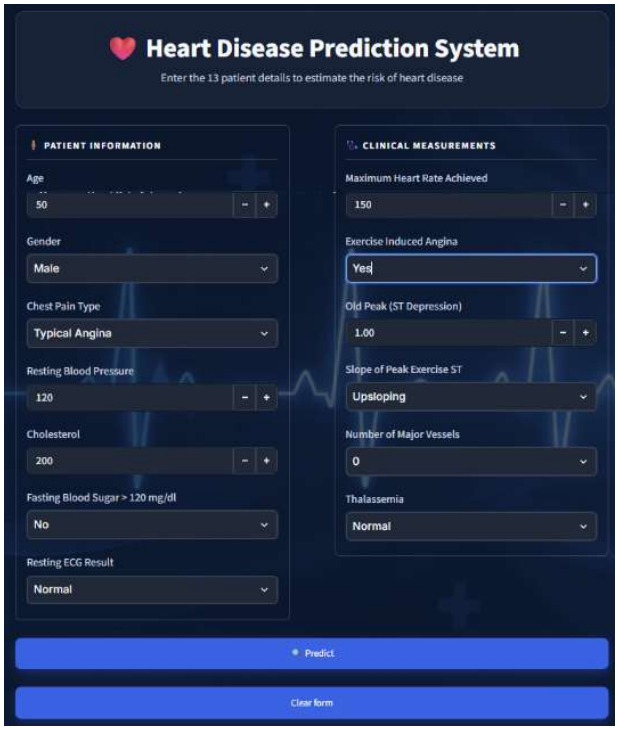


Figure 19: TC-08 – Exercise angina inputs



Figure 20: TC-08 – Exercise angina result

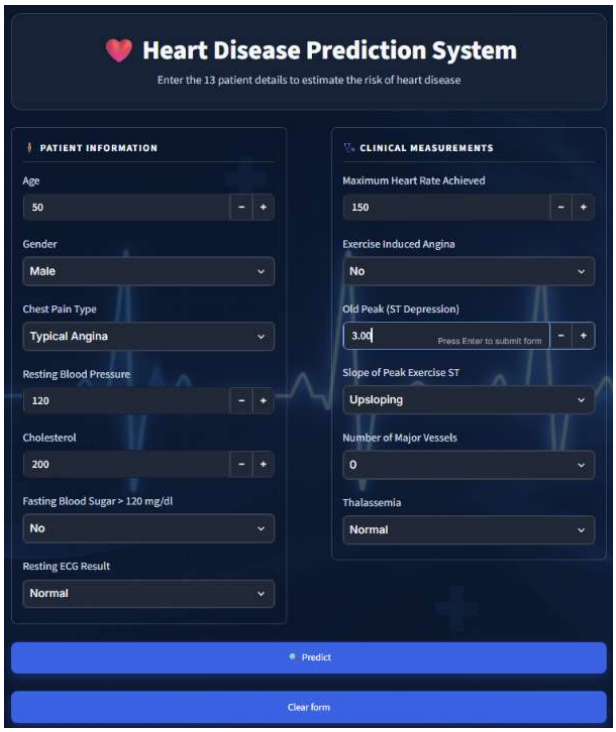


Figure 21: TC-09 – High ST depression inputs



Figure 22: TC-09 – High ST depression result

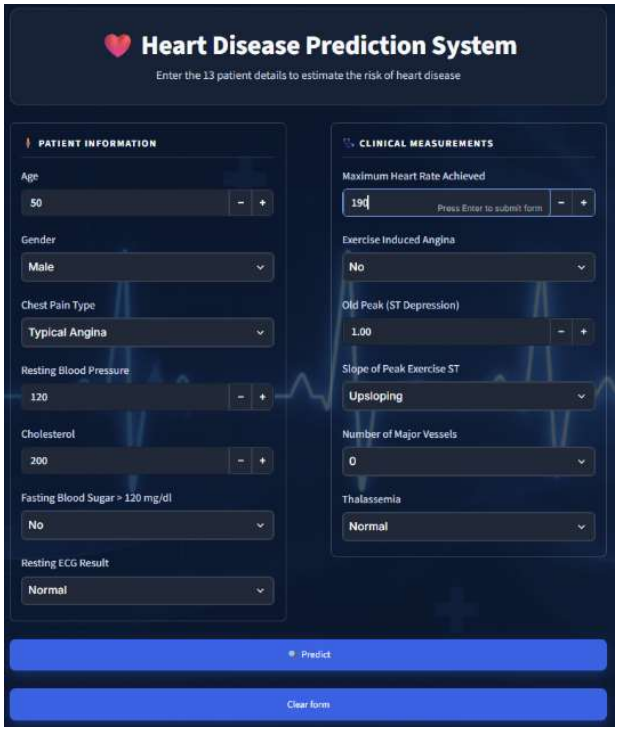


Figure 23: TC-10 – High max heart rate inputs



Figure 24: TC-10 – High max heart rate result

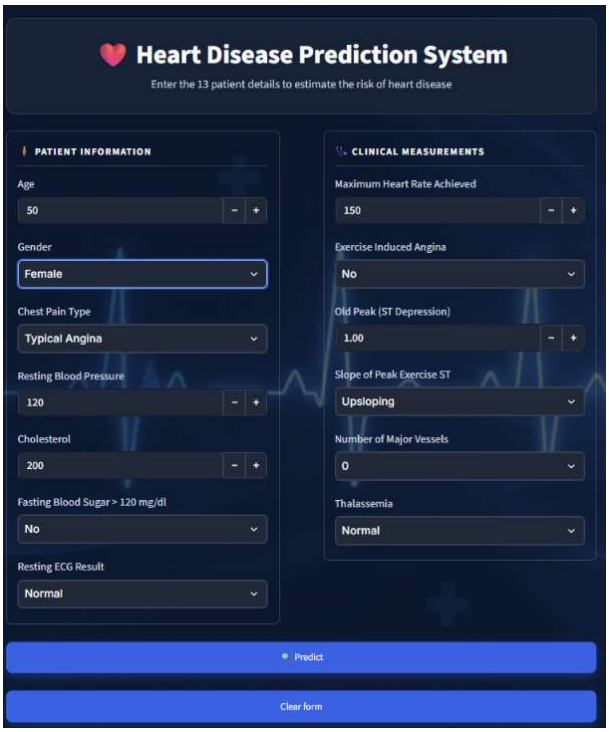


Figure 25: TC-11 – Gender change (female) inputs



Figure 26: TC-11 – Gender change (female) result

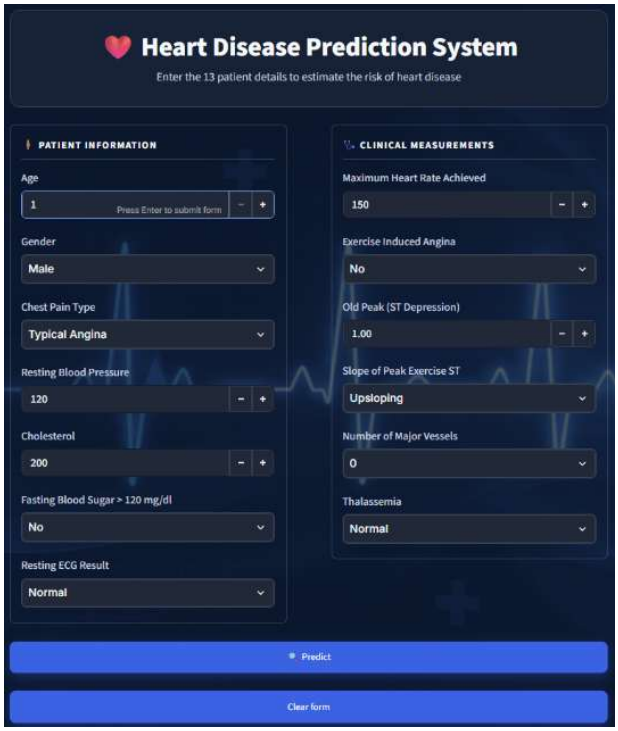


Figure 27: TC-12 – Age at minimum (Age 1) inputs



Figure 28: TC-12 – Age at minimum (Age 1) result



Figure 29: TC-13 – Repeatability, first prediction



Figure 30: TC-13 – Repeatability, second prediction

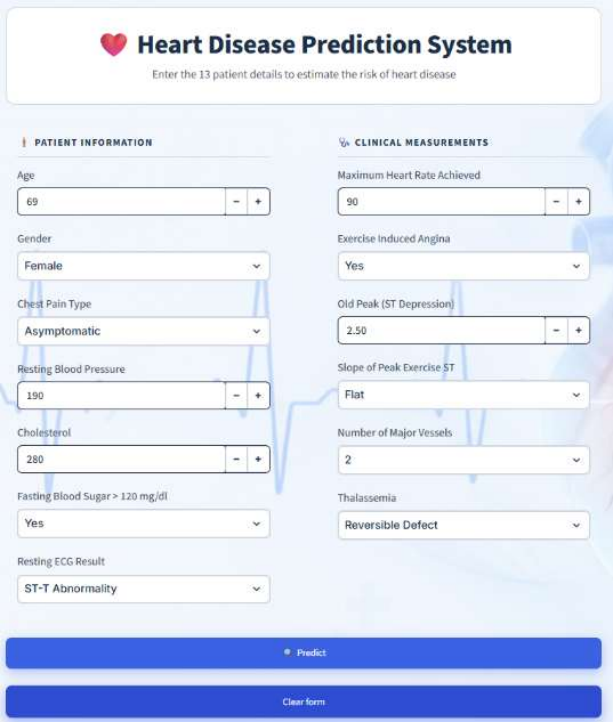


Figure 31: TC-14 – Light mode inputs

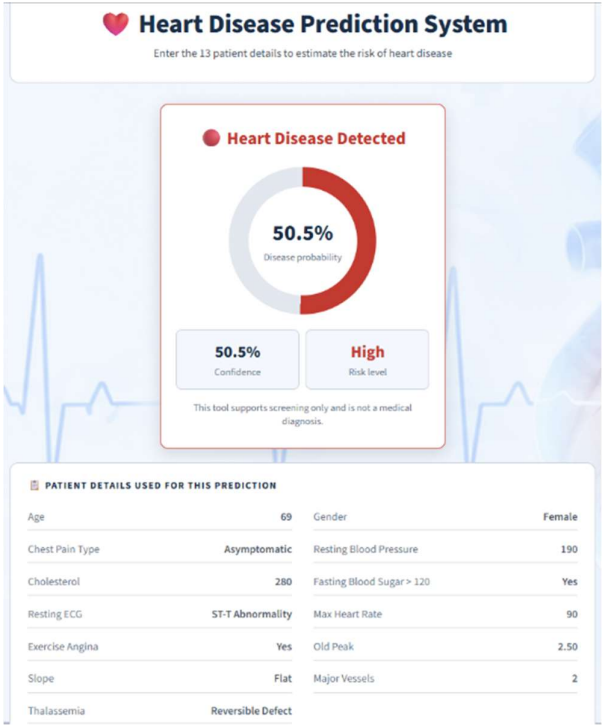


Figure 32: TC-14 – Light mode result

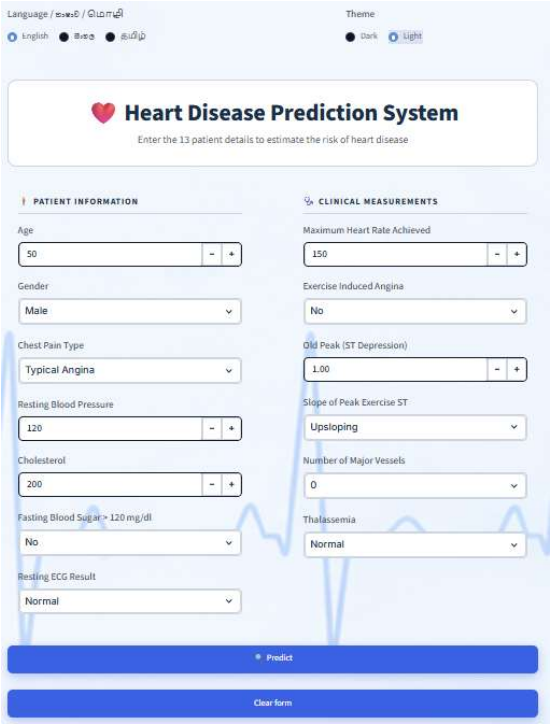


Figure 33: TC-14 – Light mode patient details

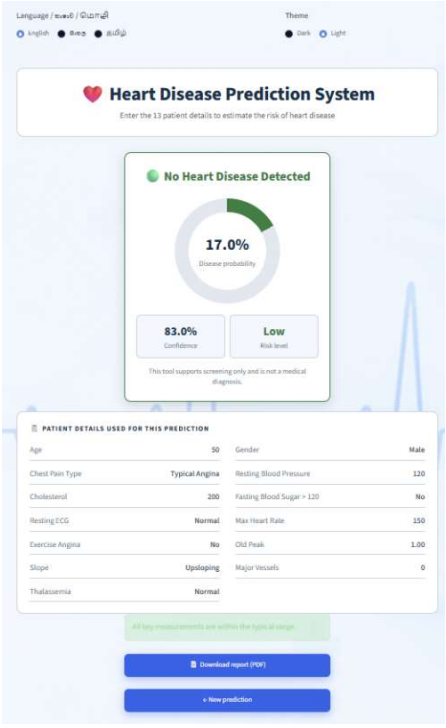


Figure 34: TC-15 – Viewed a result

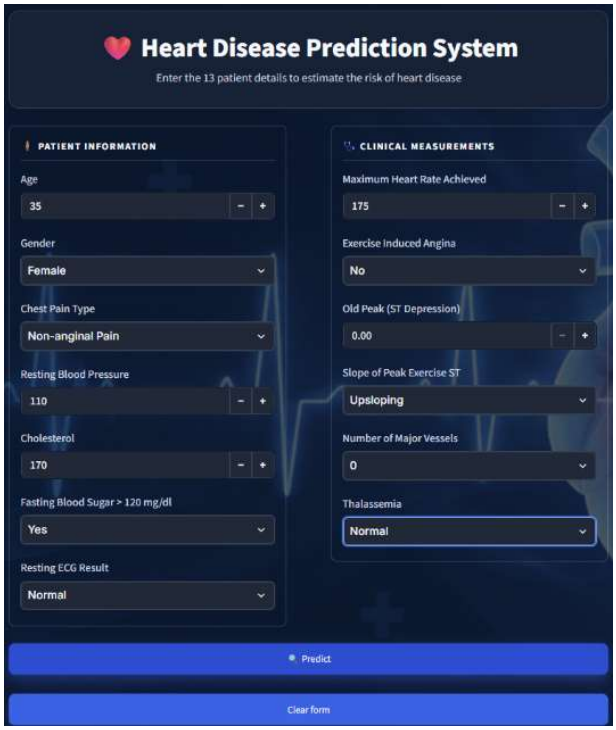


Figure 35: TC-15 – Entered a new patient (TC-03 values, but FBS = Yes)



Figure 36: TC-15 – New patient result

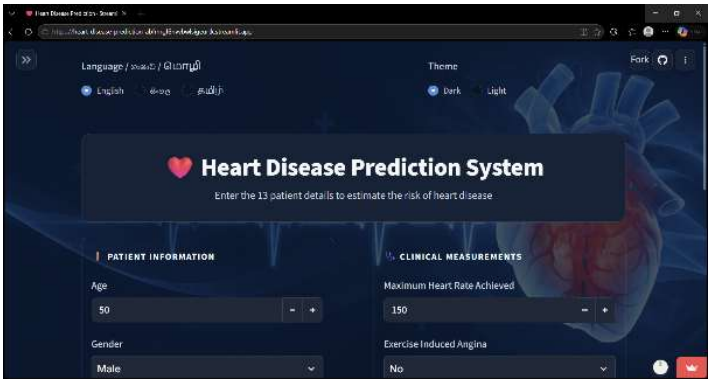


Figure 37: TC-16 – Page with sidebar closed

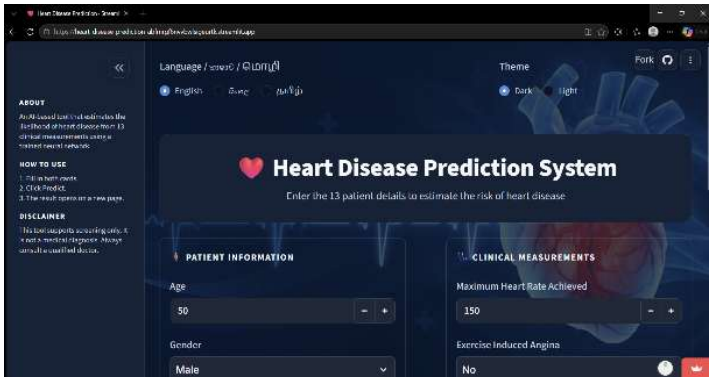


Figure 38: TC-16 – Sidebar content

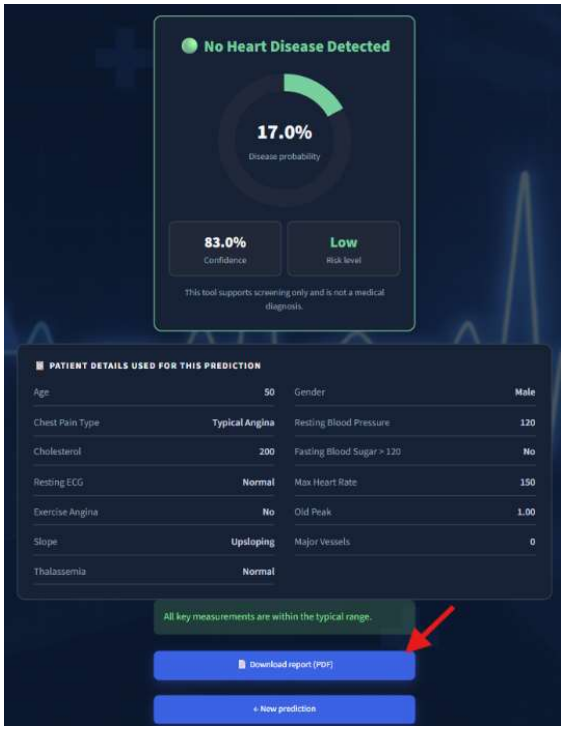


Figure 39: TC-17 – Download report button on result page

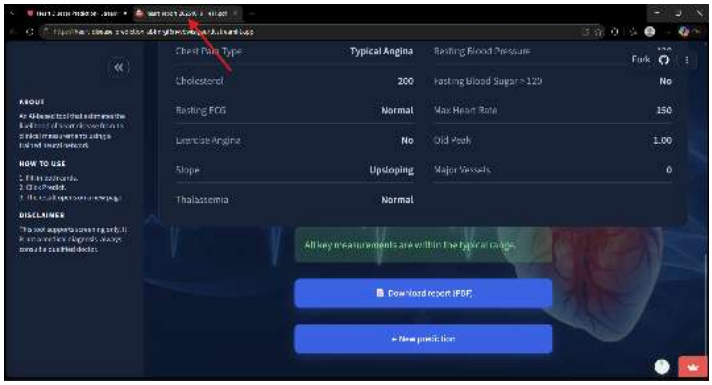


Figure 40: TC-17 – Downloaded PDF opened in a new tab



Figure 41: TC-18 – Result page (default values)

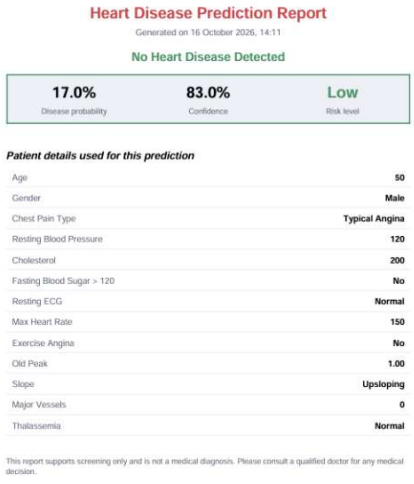


Figure 42: TC-18 – Downloaded PDF report



Figure 43: TC-19 – Language selector with three options



Figure 44: TC-20 – Sinhala input page

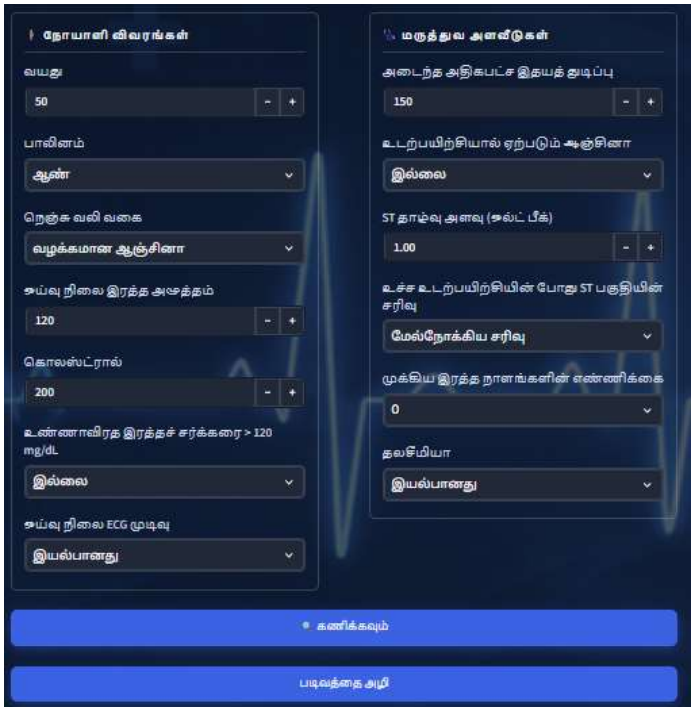


Figure 45: TC-21 – Tamil input page

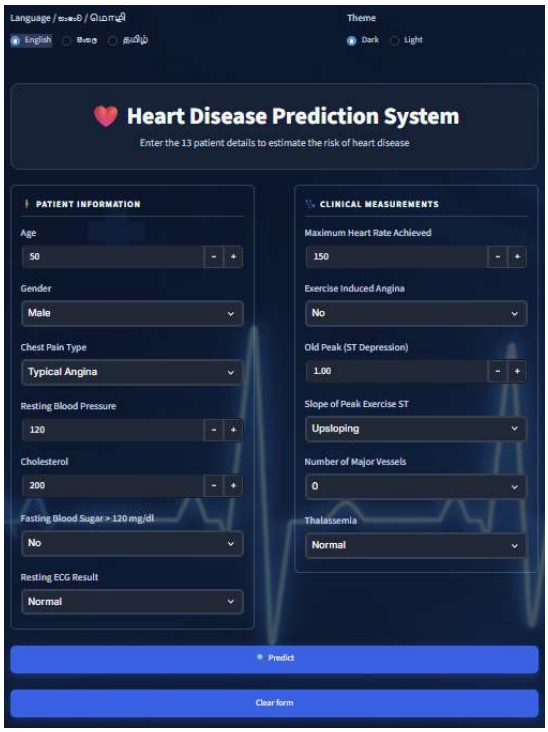


Figure 46: TC-22 – Default values in English (input page)



Figure 47: TC-22 – Default values in Sinhala (input page)

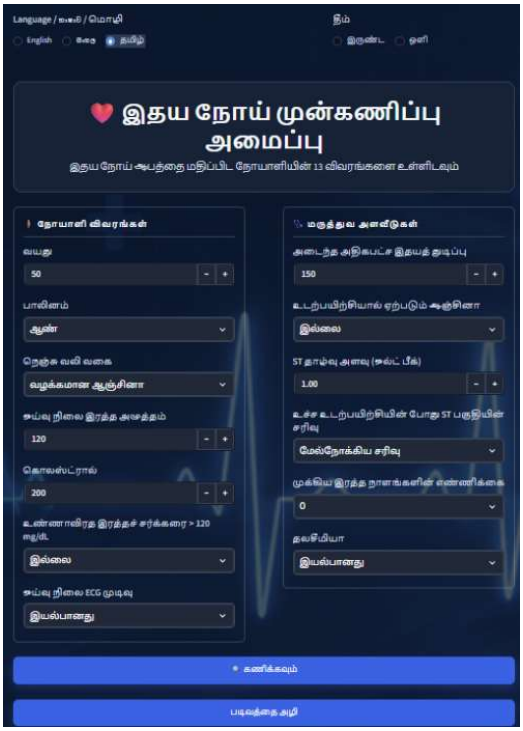


Figure 48: TC-22 – Default values in Tamil (input page)



Figure 49: TC-23 – Sinhala result page



Figure 50: TC-24 – Tamil result page

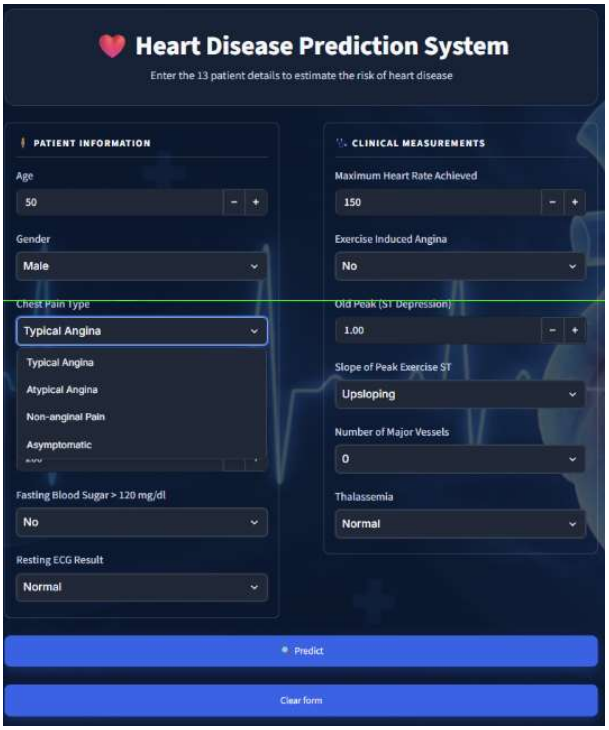


Figure 51: TC-25 – Chest pain options in English

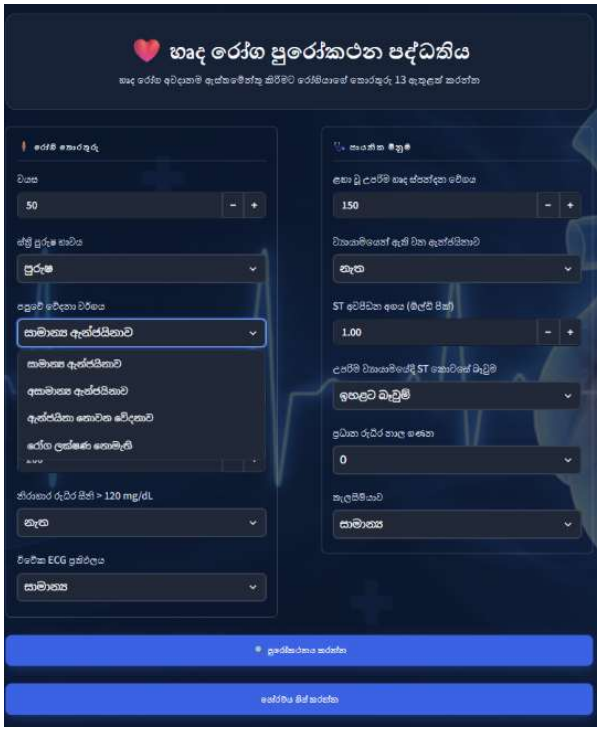


Figure 52: TC-25 – Chest pain options in Sinhala

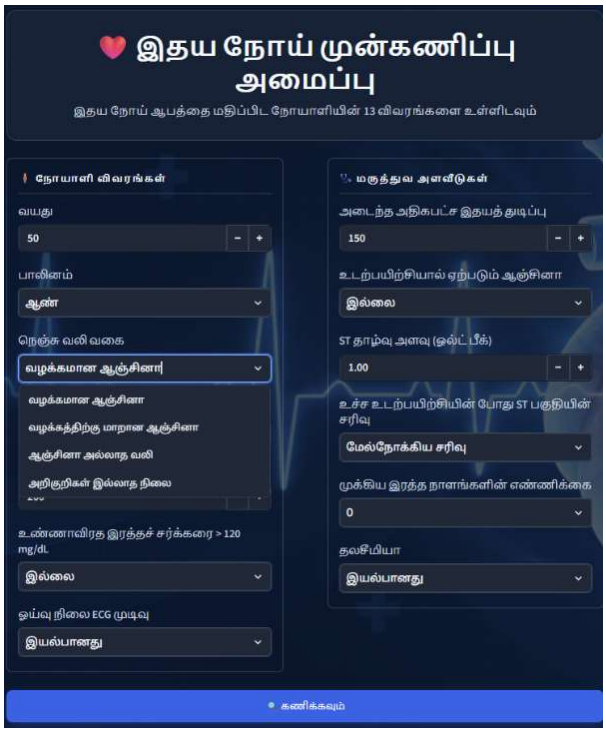


Figure 53: TC-25 – Chest pain options in Tamil



Figure 54: TC-26 – Result page in English (before changing language)



Figure 55: TC-26 – Result page after switching to Sinhala



Figure 56: TC-27 – Out-of-range warning (3 major vessels) in Sinhala



Figure 57: TC-27 – Out-of-range warning (3 major vessels) in Tamil

Chapter 6: Limitations and Discussion

6.1 Dataset Provenance and Realism

Non-integer data is recorded for some variables which should be integers (such as age with 61.46 or blood pressure with 120.52), and the diagnosis-level numbers differ from the typical 303-number Cleveland dataset [10], [11]. This means that the 1,024 number dataset must have been created through some kind of data augmentation on a smaller dataset. Performance on such data cannot be taken to be indicative of performance on real patients. It must be clearly stated where the data came from and how it was generated, and further validation of the model must be done on another real-world dataset before wider usage. Test records appear to be far away from the training data, according to a nearest-neighbour analysis (median standardized distance around 1.9), but this does not solve the provenance issue.

6.2 Input Encoding Mismatch

Chest pain is coded in the dataset as 1 to 4, ST slope as 1 to 3, and thalassemia as 3, 6, or 7. In the interface code, the inputs are mapped into chest pain 0 to 3, slope 0 to 2, and thalassemia 1 to 3, as well as providing 4 as an option for major vessels even though in the dataset this variable ranges between 0 and 3. Thus, these mappings do not correspond to the dataset codes that were used in the training, and therefore, can change predictions. For correct functioning of the interface, the mappings should be fixed to: chest pain 1 to 4, slope 1 to 3 (up, flat, down), thalassemia 3 (normal), 6 (fixed defect), 7 (reversible defect), and major vessels 0 to 3. The test set results obtained in Chapter 5 remain unchanged, since they are computed using the original dataset codes.

6.3 Model Limitations

There are only 205 samples in the test dataset, so each of the metrics is uncertain. There is only one division into train, validation, and test sets; there is no cross-validation. There is still some discrepancy in terms of accuracy between the training and validation datasets. The fixed threshold of 0.5 implies that all errors are equal, while for a screening tool, the threshold would be adjusted for more recall. Probabilities output by the neural network are not necessarily calibrated; therefore, the shown confidence is merely the model's score and not the clinical probability. The model cannot warn of inputs which are very different from those seen during training.

6.4 Scope

The model can determine whether there is disease or not in binary format; it will not be able to give details on the severity or type of the disease. The model needs only 13 predictors, and clinical test values that are not available to the general public such as fluoroscopy vessel count and thalassemia result.

6.5 Future Work

Test the model on external test data; incorporate cross-validation and parameter tuning; benchmark the results against other algorithms like logistic regression and random forest; optimize the decision threshold for maximum recall; probabilistic calibration; input range validation; and explainability (e.g., feature importance or SHAP values) to give insights into the prediction process.

Chapter 7: Ethical and Social Considerations

7.1 Data Privacy

Personal information is entered into the application. There is no storage of data in the prototype, and it is used only for calculations of a prediction. Any use requires obtaining user consent, data security, encryption, and complying with all data protection laws. The Streamlit deployment is public and is intended only for entering dummy patient data.

7.2 Bias and Fairness

The model is a reflection of the data used for training. In case the data is unrepresentative of certain gender, age range, and ethnicity, the accuracy will be lower for that group of patients, and biased data can lead to unfair outcomes in health algorithms [13]. The data used for the model training is based on a limited population, and evaluation was not conducted in subgroups, which means the problem of fairness has not been solved.

7.3 Accuracy and Reliability

With approximately 83% of accuracy, in one out of six cases the classification is wrong. The false negative outcome can lead to misdiagnosis and delayed treatment, while the false positive outcome can cause unnecessary panic. Hence, the interface indicates that the results are preliminary.

7.4 Misuse and Over-Reliance

The system might be used in place of a diagnosis, can be used by insurance companies and employers for evaluating people, and can substitute a visit to the doctor. The system can be provided to the user only as an educational and screening tool.

7.5 Human Responsibility

Diagnosis and treatment are matters of qualified doctors. The system facilitates awareness and helps the user consult the doctor but does not provide clinical decisions.

7.6 Transparency

According to the report, the dataset is augmented, the problem of interface is mentioned, and the fact that AI was used to prepare some sections of the report, as it is required in the assignment.

Chapter 8: Conclusion

In order to predict heart disease, a prototype based on ANN was developed by the team. TThrough standardizing the 13 clinical characteristics of patients and using techniques such as Batch Normalization, Dropout, early stopping, and changing the learning rate, the model obtained an accuracy of about 83% and an AUC of about 0.92 for the holdout set. It serves as an example of how supervised classification neural networks may translate patient characteristics into a quick initial risk assessment provided via a convenient interface. Its importance lies in the fact that it demonstrates the potential: the dataset seems to be expanded, the testing is performed only once on a very small part of the whole dataset, and the encoding for the interface should be aligned with the training set before getting any reasonable results.

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Individual Contribution

Student Name	KLAN Shehara	**Registration No.**	D/BSE/25/0017
Degree / Intake	BSc (Hons) in Software Engineering Intake 42	**Group No.**	42
Project Title	AI-Based Heart Disease Risk Prediction System		

Technical Contributions

Table 10: Technical contributions – KLAN Shehara

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
ANN Architecture Design	Designed the Artificial Neural Network architecture to process 13 patient-related input features and perform binary classification.	Artificial Neural Network (ANN), TensorFlow, Keras	ANN model architecture and model implementation code
ANN Implementation	Implemented the neural network using hidden layers with ReLU activation and an output layer with sigmoid activation to predict heart disease.	Python, TensorFlow, Keras, ReLU, Sigmoid	Model-building code and heart_model.keras file
Model Training	Trained the ANN using the prepared dataset and monitored training and validation performance to evaluate the model's learning progress.	TensorFlow, Keras	Training code, training history
Training Optimization	Applied Early Stopping and learning-rate adjustment to improve the training process and help reduce overfitting.	EarlyStopping, ReduceLROnPlateau	Training configuration and validation results
Model Evaluation and Performance Analysis	Evaluated the trained ANN model using multiple performance metrics and analyzed the results to assess its predictive performance, strengths, and limitations in predicting heart disease.	Python, Scikit-learn, Accuracy, Precision, Recall, F1-score, AUC	Classification report and confusion matrix
Project Documentation	Participated in preparing and updating project documentation by contributing information about ANN architecture, model training, training optimization, and model performance evaluation.	Microsoft Word, project results, ANN implementation details	Project proposal, progress reports, and project documentation sections contributed by me

Evidence

1. ANN model architecture and model implementation code

```
# ============================================================
# 8. BUILD NEURAL NETWORK
# ============================================================

input_features = X_train_processed.shape[1]
model = Sequential([
    Input(shape=(input_features,)),
    Dense(64, activation="relu"),
    BatchNormalization(),
    Dropout(0.20),
    Dense(32, activation="relu"),
    BatchNormalization(),
    Dropout(0.15),
    Dense(16, activation="relu"),
    Dense(1, activation="sigmoid")
])
```

Figure 58: Neural network architecture code (Sequential model)

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 64)	896
batch_normalization (BatchNormalization)	(None, 64)	256
dropout (Dropout)	(None, 64)	0
dense_1 (Dense)	(None, 32)	2,080
batch_normalization_1 (BatchNormalization)	(None, 32)	128
dropout_1 (Dropout)	(None, 32)	0
dense_2 (Dense)	(None, 16)	528
dense_3 (Dense)	(None, 1)	17

Total params: 3,905 (15.25 KB)
Trainable params: 3,713 (14.50 KB)
Non-trainable params: 192 (768.00 B)

Figure 59: Model summary showing layers, output shapes and parameter counts

2. Model-building code and heart_model.keras file



Figure 60: Saved model file (heart_model.keras) in the project folder

```
# ============================================================
# 9. COMPILE MODEL
# ============================================================

model.compile(
    optimizer=Adam(
        learning_rate=0.001
    ),
    loss="binary_crossentropy",
    metrics=[
        "accuracy",
        tf.keras.metrics.AUC(
            name="auc"
        ),
        tf.keras.metrics.Precision(
            name="precision"
        ),
        tf.keras.metrics.Recall(
            name="recall"
        )
    ]
)
print("\nModel compiled successfully!")
model.summary()
```

Figure 61: Model compilation code (Adam optimizer, binary cross-entropy loss and metrics)

3. Training code and training history

```
# ============================================================
# 11. TRAIN NEURAL NETWORK
# ============================================================

print("\n========== Training Neural Network ==========")
history = model.fit(
    X_train_processed,
    y_train,
    validation_data=(
        X_val_processed,
        y_val
    ),
    epochs=150,
    batch_size=32,
    callbacks=[
        early_stopping,
        reduce_lr
    ],
    verbose=1
)
```

Figure 62: Model training code (150 epochs, batch size 32, with callbacks)

```
========== Training Neural Network ==========
Epoch 1/150
 1/21 ━━━━━━━━━━━━━━━━━━━━ 38s 2s/step - accuracy: 0.5000 - auc: 0.5830 - loss: 0.8614 - precision: 0.8000 - recall: 0
21/21 ━━━━━━━━━━━━━━━━━━━━ 2s 19ms/step - accuracy: 0.6092 - auc: 0.6743 - loss: 0.6649 - precision: 0.6531 - recall:
0.3189 - val_accuracy: 0.6524 - val_auc: 0.8515 - val_loss: 0.6046 - val_precision: 0.8750 - val_recall: 0.2800 - lear
ning_rate: 0.0010
Epoch 2/150
 1/21 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - accuracy: 0.7812 - auc: 1.0000 - loss: 0.4456 - precision: 1.0000 - recall:
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7649 - auc: 0.8822 - loss: 0.4737 - precision: 0.8182 - recall: 0
.6279 - val_accuracy: 0.8354 - val_auc: 0.9121 - val_loss: 0.5193 - val_precision: 0.8333 - val_recall: 0.8000 - learn
ing_rate: 0.0010
Epoch 3/150
 1/21 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - accuracy: 0.8438 - auc: 1.0000 - loss: 0.3684 - precision: 1.0000 - recall:
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8382 - auc: 0.9065 - loss: 0.4123 - precision: 0.8283 - recall: 0
.8173 - val_accuracy: 0.8537 - val_auc: 0.9267 - val_loss: 0.4468 - val_precision: 0.8312 - val_recall: 0.8533 - learn
ing_rate: 0.0010
Epoch 4/150
 1/21 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - accuracy: 0.9688 - auc: 0.9919 - loss: 0.2789 - precision: 1.0000 - recall:
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8427 - auc: 0.9187 - loss: 0.3710 - precision: 0.8094 - recall: 0
.8605 - val_accuracy: 0.8659 - val_auc: 0.9368 - val_loss: 0.3919 - val_precision: 0.8354 - val_recall: 0.8800 - learn
ing_rate: 0.0010
Epoch 5/150
 1/21 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - accuracy: 0.9375 - auc: 0.9960 - loss: 0.2443 - precision: 0.9474 - recall:
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8672 - auc: 0.9296 - loss: 0.3414 - precision: 0.8344 - recall: 0
.8870 - val_accuracy: 0.8841 - val_auc: 0.9408 - val_loss: 0.3575 - val_precision: 0.8500 - val_recall: 0.9067 - learn
ing_rate: 0.0010
Epoch 6/150
 1/21 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - accuracy: 0.9688 - auc: 1.0000 - loss: 0.1613 - precision: 1.0000 - recall:
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8809 - auc: 0.9466 - loss: 0.3054 - precision: 0.8389 - recall: 0
.9169 - val_accuracy: 0.8780 - val_auc: 0.9436 - val_loss: 0.3359 - val_precision: 0.8481 - val_recall: 0.8933 - learn
ing_rate: 0.0010
Epoch 7/150
 1/21 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.1508 - precision: 1.0000 - recall:
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8779 - auc: 0.9476 - loss: 0.2953 - precision: 0.8464 - recall: 0
.8970 - val_accuracy: 0.8659 - val_auc: 0.9446 - val_loss: 0.3194 - val_precision: 0.8272 - val_recall: 0.8933 - learn
ing_rate: 0.0010
Epoch 8/150
```

Figure 63: Training log – epochs 1 to 8

```
PS C:\Users\Shehara\Downloads\HeartDiseasePrediction-main> python main.py
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8977 - auc: 0.9691 - loss: 0.2238 - precision: 0.8824 - recall: 0
.8970 - val_accuracy: 0.8841 - val_auc: 0.9447 - val_loss: 0.3083 - val_precision: 0.8590 - val_recall: 0.8933 - learn
ing_rate: 0.0010
Epoch 16/150
 1/21 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - accuracy: 0.9688 - auc: 0.9960 - loss: 0.1334 - precision: 1.0000 - recall:
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9084 - auc: 0.9672 - loss: 0.2317 - precision: 0.9030 - recall: 0
.8970 - val_accuracy: 0.8780 - val_auc: 0.9448 - val_loss: 0.3113 - val_precision: 0.8571 - val_recall: 0.8800 - learn
ing_rate: 0.0010
Epoch 17/150
 1/21 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - accuracy: 0.9688 - auc: 0.9960 - loss: 0.1258 - precision: 1.0000 - recall:
0.9474
Epoch 17: ReduceLROnPlateau reducing learning rate to 0.0005000000237487257.
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8947 - auc: 0.9686 - loss: 0.2275 - precision: 0.8816 - recall: 0
.8904 - val_accuracy: 0.8780 - val_auc: 0.9437 - val_loss: 0.3148 - val_precision: 0.8571 - val_recall: 0.8800 - learn
ing_rate: 0.0010
Epoch 18/150
 1/21 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0855 - precision: 1.0000 - recall:
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9084 - auc: 0.9721 - loss: 0.2143 - precision: 0.9003 - recall: 0
.9003 - val_accuracy: 0.8841 - val_auc: 0.9442 - val_loss: 0.3159 - val_precision: 0.8590 - val_recall: 0.8933 - learn
ing_rate: 5.0000e-04
Epoch 19/150
 1/21 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - accuracy: 0.9688 - auc: 1.0000 - loss: 0.1231 - precision: 1.0000 - recall:
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9130 - auc: 0.9688 - loss: 0.2267 - precision: 0.8987 - recall: 0
.9136 - val_accuracy: 0.8963 - val_auc: 0.9443 - val_loss: 0.3143 - val_precision: 0.8718 - val_recall: 0.9067 - learn
ing_rate: 5.0000e-04
Epoch 20/150
 1/21 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - accuracy: 0.9375 - auc: 1.0000 - loss: 0.1365 - precision: 1.0000 - recall:
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8992 - auc: 0.9725 - loss: 0.2130 - precision: 0.8852 - recall: 0
.8970 - val_accuracy: 0.8963 - val_auc: 0.9446 - val_loss: 0.3120 - val_precision: 0.8718 - val_recall: 0.9067 - learn
ing_rate: 5.0000e-04
Epoch 21/150
 1/21 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - accuracy: 0.9688 - auc: 0.9960 - loss: 0.1481 - precision: 1.0000 - recall:
21/21 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9191 - auc: 0.9731 - loss: 0.2098 - precision: 0.9026 - recall: 0
.9236 - val_accuracy: 0.8902 - val_auc: 0.9459 - val_loss: 0.3107 - val_precision: 0.8701 - val_recall: 0.8933 - learn
ing_rate: 5.0000e-04
Epoch 22/150
 1/21 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - accuracy: 0.9375 - auc: 0.9960 - loss: 0.1201 - precision: 1.0000 - recall:
0.8947
```

Figure 64: Training log – epochs 16 to 22 (learning-rate reduction at epoch 17)

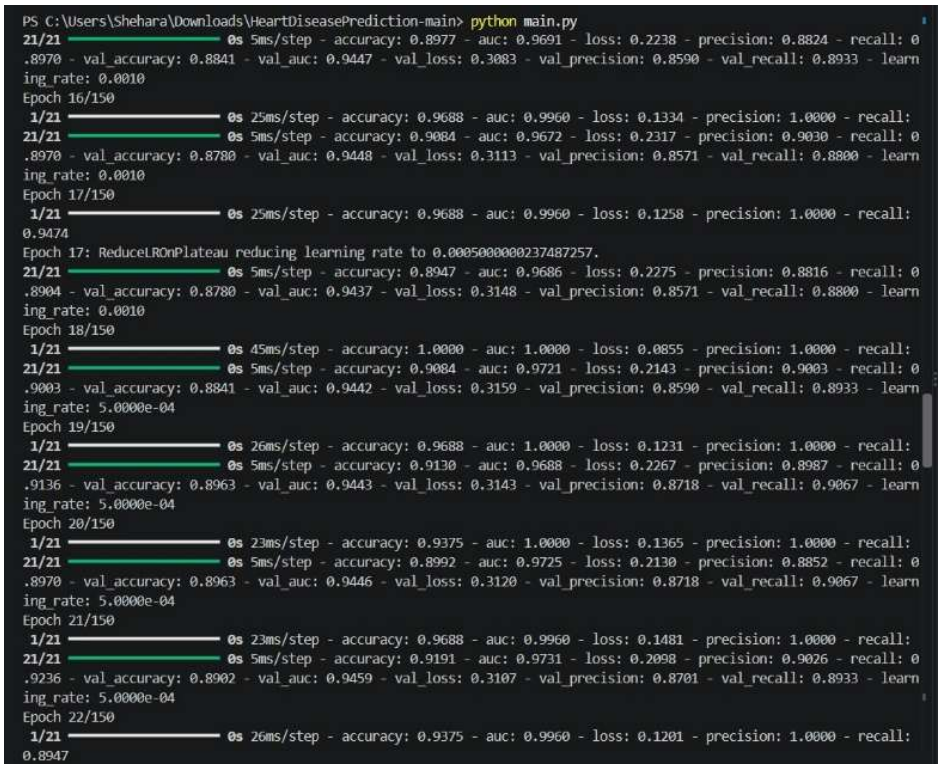


Figure 65: Training log – epochs 16 to 22 (second capture)



Figure 66: Training log – epochs 24 to 30

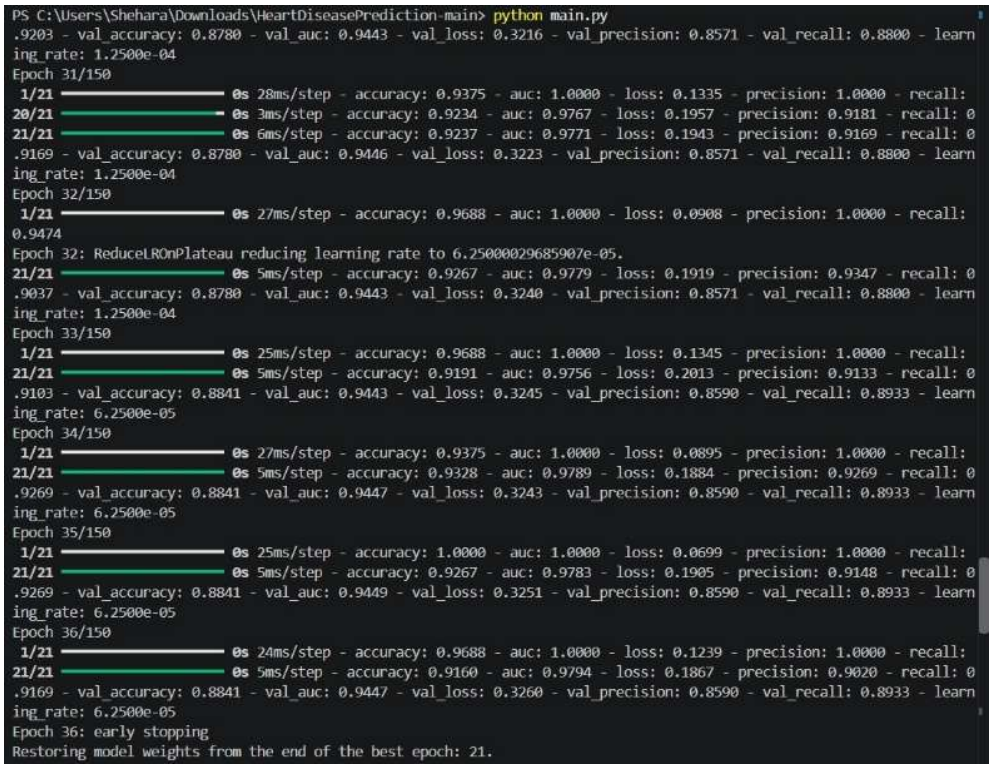


Figure 67: Training log – epochs 31 to 36 (early stopping)

4. Training configuration and validation results

```
# ============================================================
# 10. CALLBACKS
# ============================================================

early_stopping = EarlyStopping(
    monitor="val_auc",
    mode="max",
    patience=15,
    restore_best_weights=True,
    verbose=1
)
reduce_lr = ReduceLROnPlateau(
    monitor="val_loss",
    factor=0.5,
    patience=5,
    min_lr=0.00001,
    verbose=1
)
```

Figure 68: Training callbacks code (EarlyStopping and ReduceLROnPlateau)

```
========== Final Training Results ==========
Training Accuracy: 0.916
Validation Accuracy: 0.8841
Training AUC: 0.9794
Validation AUC: 0.9447
Training Loss: 0.1867
Validation Loss: 0.326
Number of Epochs Completed: 36

Scaler saved successfully!
Model saved successfully!
```

Figure 69: Final training results

5. Classification report and confusion matrix

```
========== Confusion Matrix ==========
[[97 14]
 [16 78]]

========== Classification Report ==========
                precision    recall  f1-score   support

No Heart Disease       0.86      0.87      0.87       111
   Heart Disease       0.85      0.83      0.84        94

        accuracy                           0.85       205
       macro avg       0.85      0.85      0.85       205
    weighted avg       0.85      0.85      0.85       205
```

Figure 70: Confusion matrix and classification report on the test set

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:		**Date:**	17/10/2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.

Student Name	Wandana Gunasekara	Registration No.	D/COE/25/0001
Degree / Intake	BSc (Hons) in Computer Engineering Intake 42	Group No.	42
Project Title	AI-Based Heart Disease Risk Prediction System		

Technical Contributions

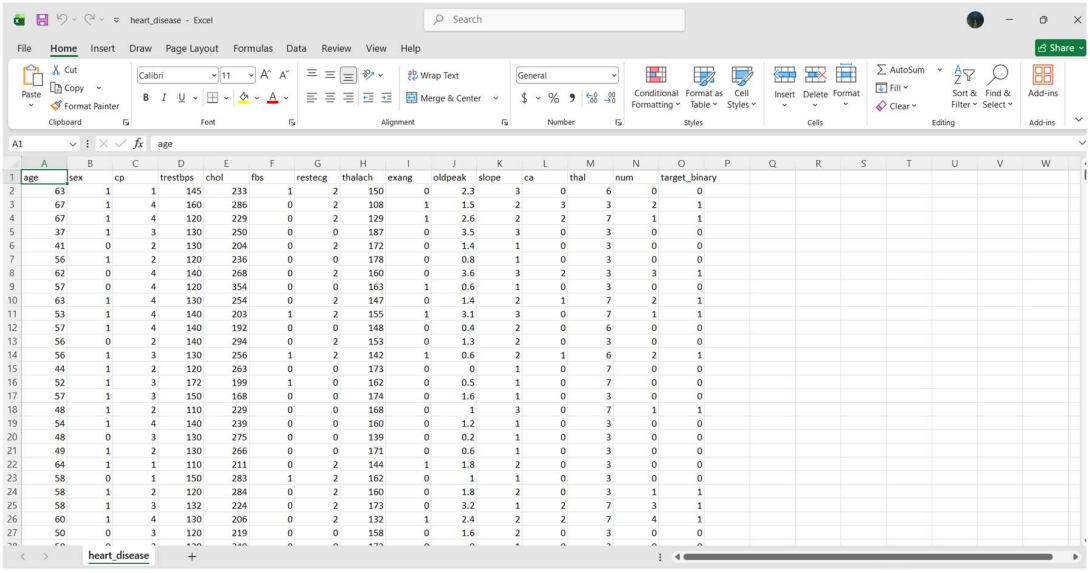
Table 11: Technical contributions – Wandana Gunasekara

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Dataset Selection, Data Quality and Feature Analysis	Researched and selected the heart disease dataset (13 clinical attributes, 1,024 records). Checked for missing values and duplicates and analysed the target distribution (554 no disease / 470 disease), so no resampling was needed. Studied the meaning of the 13 features and noted that the dataset is larger than the original 303-record Cleveland data and has decimal values in integer fields, so it was likely augmented.	Python, pandas; class distribution analysis; UCI Heart Disease (Cleveland) documentation	heart_disease.csv; data quality output (shape, missing values, duplicates, class counts); dataset sections of the Final Report
Literature and Background Research	Researched sources on heart disease prediction with machine learning, deep neural networks, feature selection, the Adam optimizer and ethical issues such as algorithmic bias in healthcare.	Zotero (reference management); academic and technical sources	Research background and ethical considerations sections of the Final Report
Multilingual Interface and Themes	Added multilingual support (English, Sinhala, Tamil) and light/dark themes with background images and a language/theme bar to the web app. Later refined these features and fixed the sidebar.	Python, Streamlit, custom CSS, Google Fonts (Inter, Noto Sans Sinhala/Tamil)	app.py, backgrounds.py, style.css, dark_bg.png, light_bg.png; commits 334932e, da36dad, 7c85417, 2ce6029; screenshots
Downloadable PDF Prediction Report	Implemented a downloadable PDF report of the patient's prediction result and later updated the report generation.	Python, ReportLab, Streamlit	app.py, requirements.txt; commits 229a485, e3e0127; sample PDF report screenshot
Responsive Design and Final System Integration	Improved the web app and desktop GUI layout for mobile, tablet and desktop screens. Finalized the system by integrating the trained TensorFlow/Keras model with both the web app and the desktop GUI so that they give consistent predictions.	Streamlit, Tkinter, CSS media queries, TensorFlow/Keras	app.py, gui.py

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
System Evaluation and Testing	Evaluated the completed system by entering a range of sample patient inputs into the web app and analysing the predicted risk results to check that the outputs were sensible and consistent.	Streamlit web app; manual test cases; TensorFlow/Keras model predictions	Sample input and result screenshots; results
Repository Maintenance and Merging	Resolved merge conflicts, fixed requirements.txt encoding to UTF-8, merged teammates' pull requests and the language/theme branch, and removed unused old files.	Git, GitHub	Commits e94edaf, e253f4e, 62d5037, acde2ba, f8845d8, cfdc86f; GitHub commit history screenshot
Final Report and Project Proposal Documentation	Compiled the complete Final Report and co-prepared the Project Proposal with all group members.	Microsoft Word; technical writing	Final Report and Project Proposal documents

Evidence

1. Dataset Selection, Data Quality and Feature Analysis



age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	num	target_binary
63	1	1	145	233	1	2	150	0	2.3	3	0	6	0	0
67	1	4	160	286	0	2	108	1	1.5	2	3	3	2	1
67	1	4	120	229	0	2	129	1	2.6	2	2	7	1	1
37	1	3	130	250	0	0	187	0	3.5	3	0	3	0	0
41	0	2	130	204	0	2	172	0	1.4	1	0	3	0	0
56	1	2	120	236	0	0	178	0	0.8	1	0	3	0	0
62	0	4	140	268	0	2	160	0	3.6	3	2	3	3	1
57	0	4	120	354	0	0	163	1	0.6	1	0	3	0	0
63	1	4	130	254	0	2	147	0	1.4	2	1	7	2	1
53	1	4	140	203	1	2	155	1	3.1	3	0	7	1	1
57	1	4	140	192	0	0	148	0	0.4	2	0	6	0	0
56	0	2	140	294	0	2	153	0	1.3	2	0	3	0	0
56	1	3	130	256	1	2	142	1	0.6	2	1	6	2	1
44	1	2	120	263	0	0	173	0	0	1	0	7	0	0
52	1	3	172	199	1	0	162	0	0.5	1	0	7	0	0
57	1	3	150	168	0	0	174	0	1.6	1	0	3	0	0
48	1	2	110	229	0	0	168	0	1	3	0	7	1	1
54	1	4	140	239	0	0	160	0	1.2	1	0	3	0	0
48	0	3	130	275	0	0	139	0	0.2	1	0	3	0	0
49	1	2	130	266	0	0	171	0	0.6	1	0	3	0	0
64	1	1	110	211	0	2	144	1	1.8	2	0	3	0	0
58	0	1	150	283	1	2	162	0	1	1	0	3	0	0
58	1	2	120	284	0	2	160	0	1.8	2	0	3	1	1
58	1	3	132	224	0	2	173	0	3.2	1	2	7	3	1
60	1	4	130	206	0	2	132	1	2.4	2	2	7	4	1
50	0	3	120	219	0	0	158	0	1.6	2	0	3	0	0

Figure 71: heart_disease.csv opened in Excel (header row and first rows showing the 13 features and target)

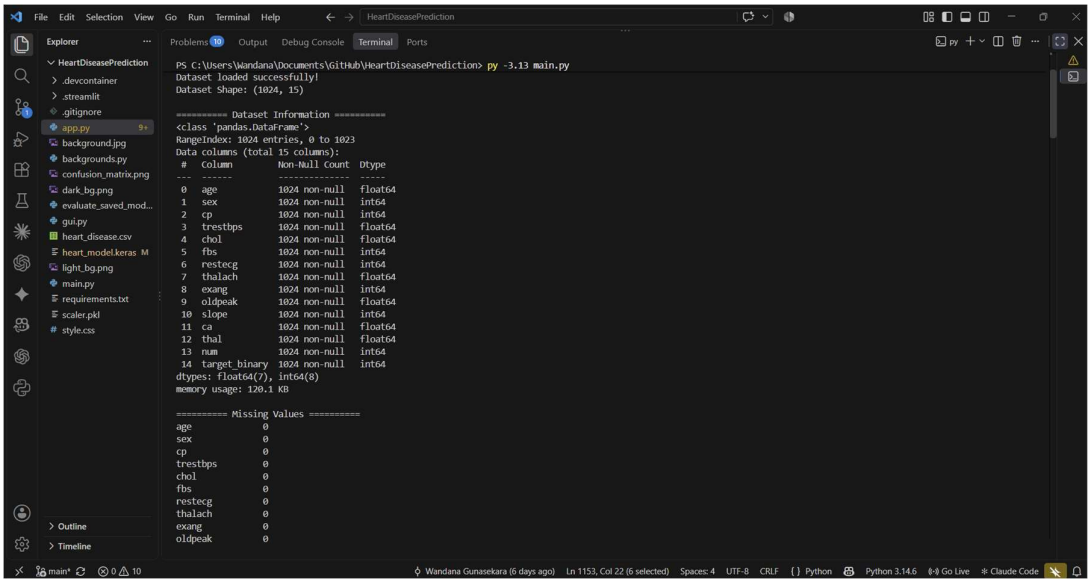


Figure 72: Python output of the dataset (shape, missing values, duplicates and class counts)

2. Literature and Background Research

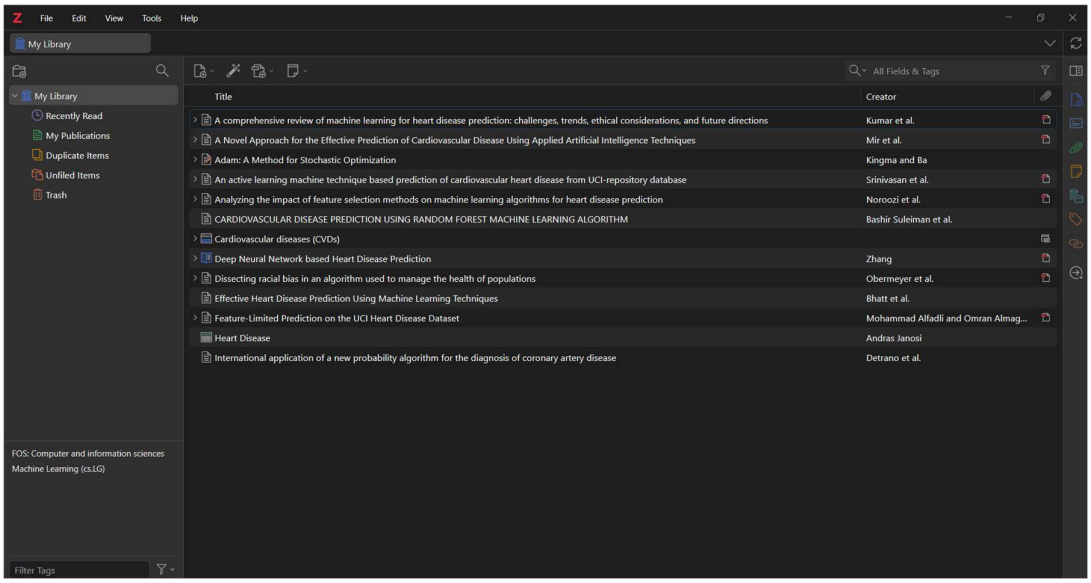


Figure 73: Research papers collected in Zotero

3. Multilingual Interface and Themes

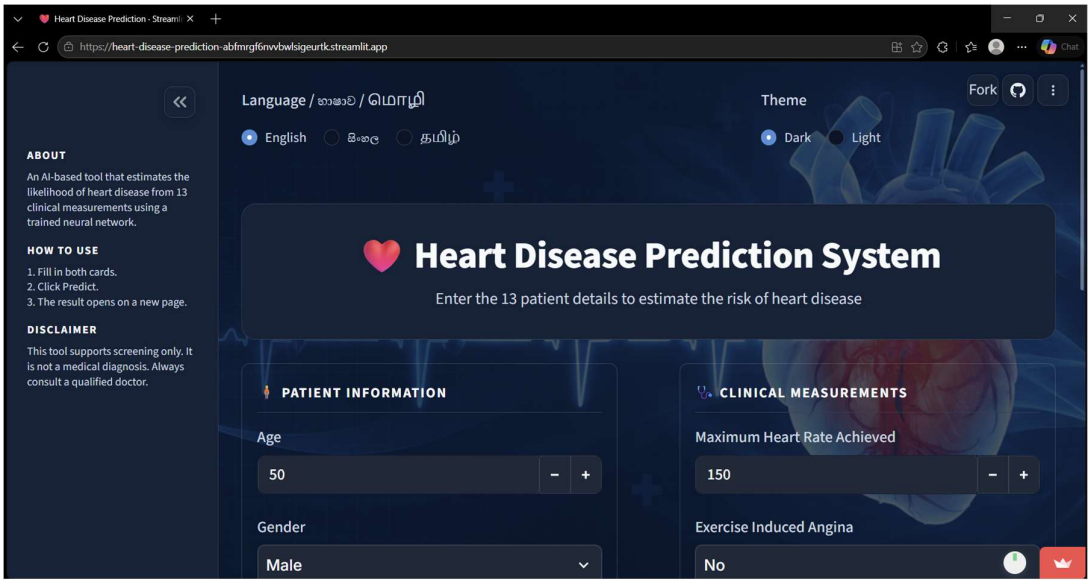


Figure 74: Web app in English, Sinhala and Tamil (language bar visible)

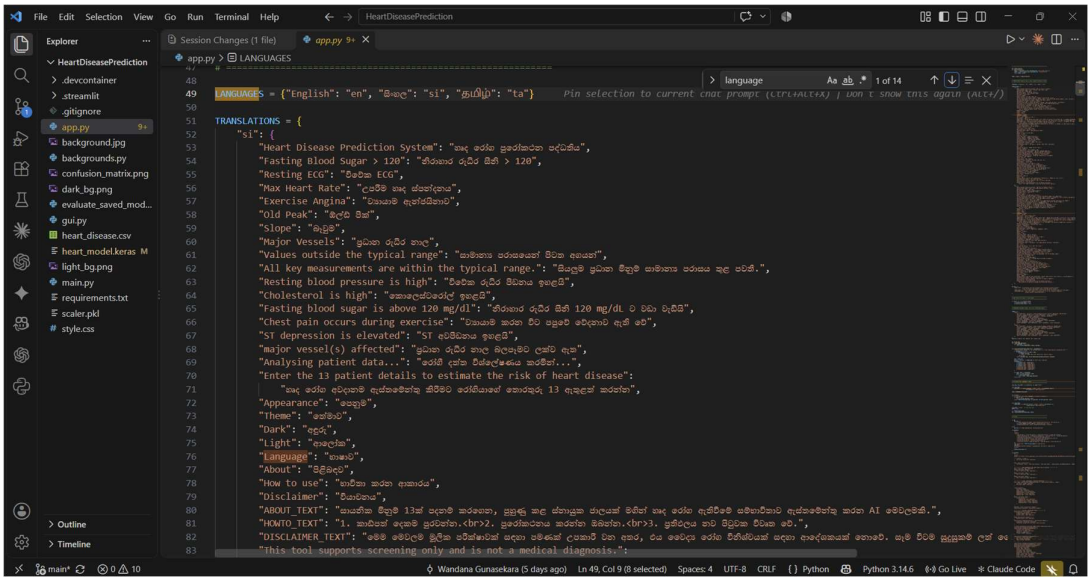


Figure 75: Language translation dictionary in app.py

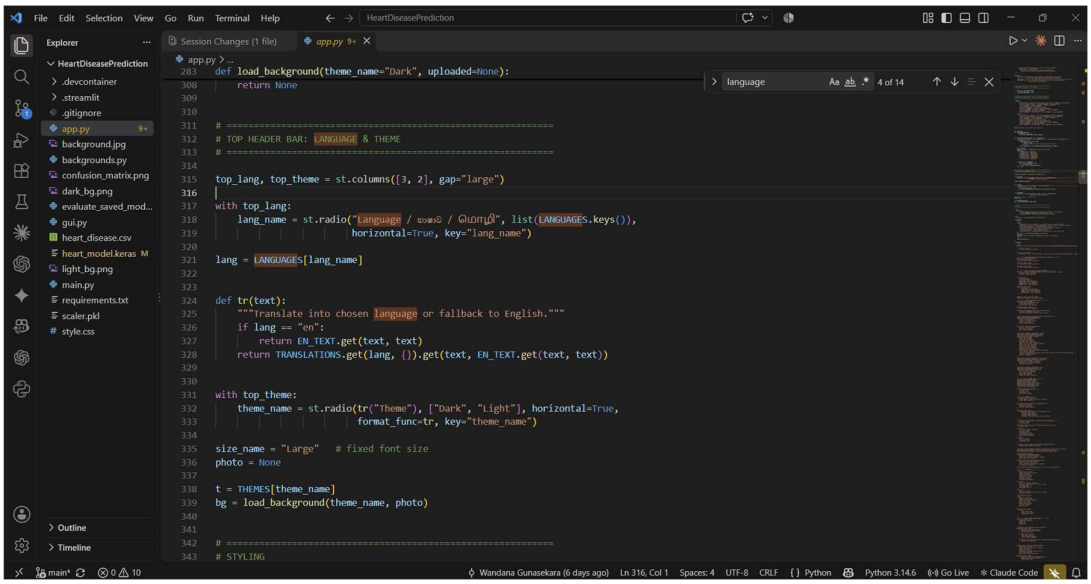


Figure 76: Language and theme bar code in app.py

4. Downloadable PDF Prediction Report

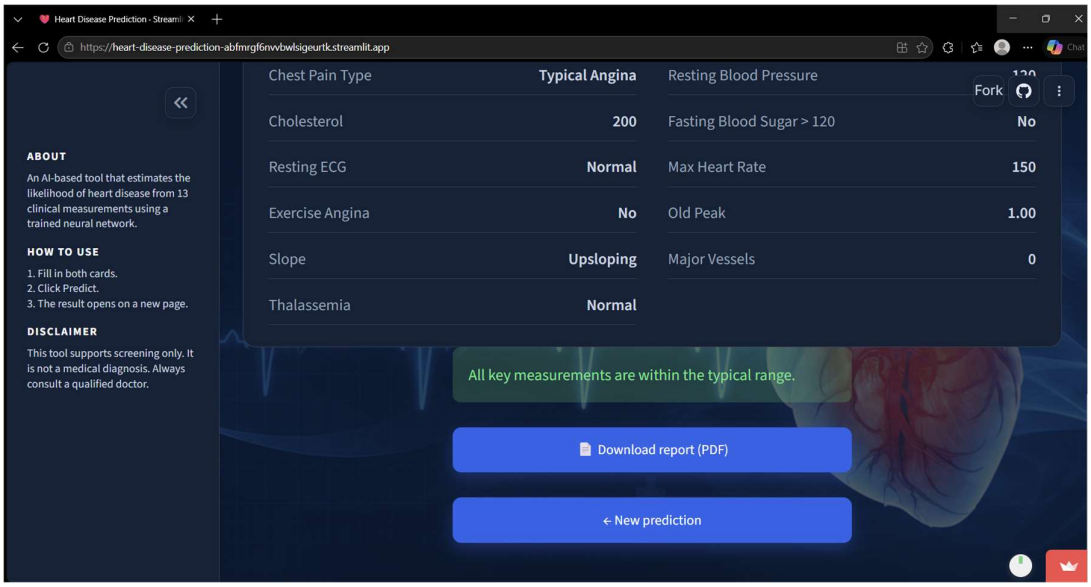


Figure 77: Result page showing the "Download PDF report" button

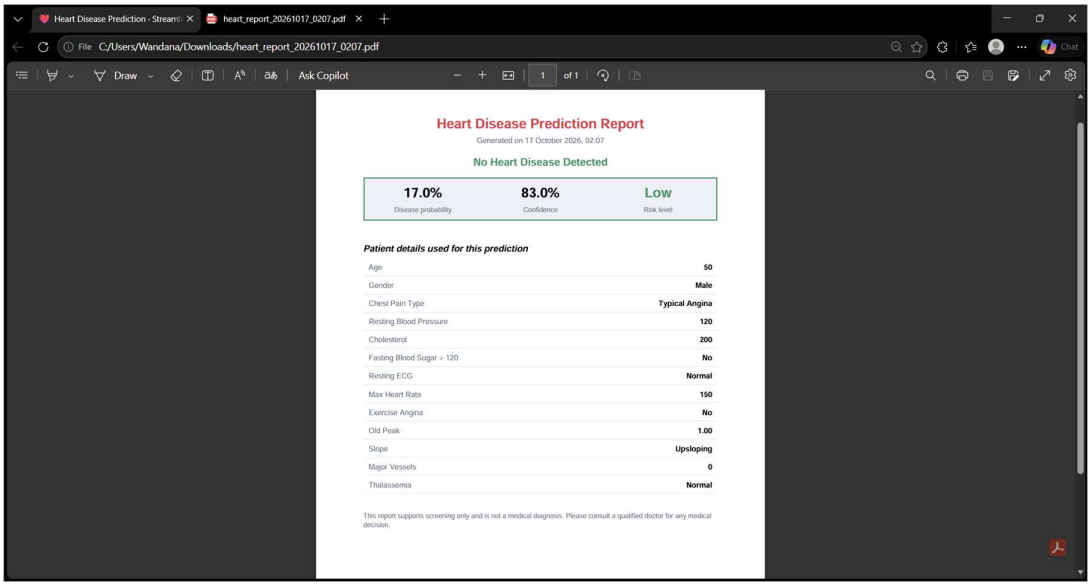


Figure 78: The generated PDF report opened

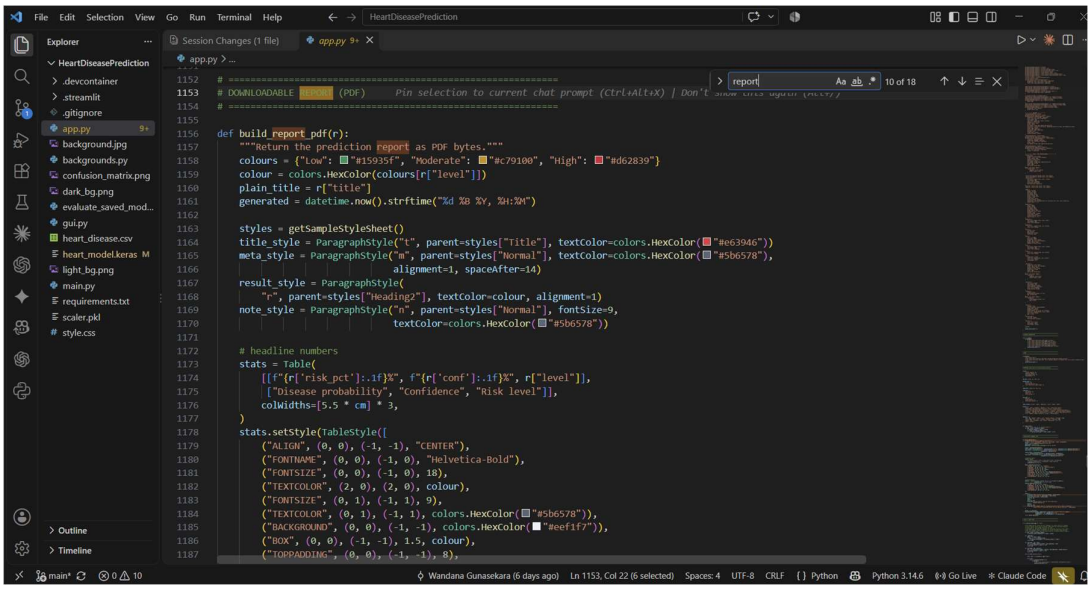


Figure 79: ReportLab code in app.py

5. Responsive Design and Final System Integration

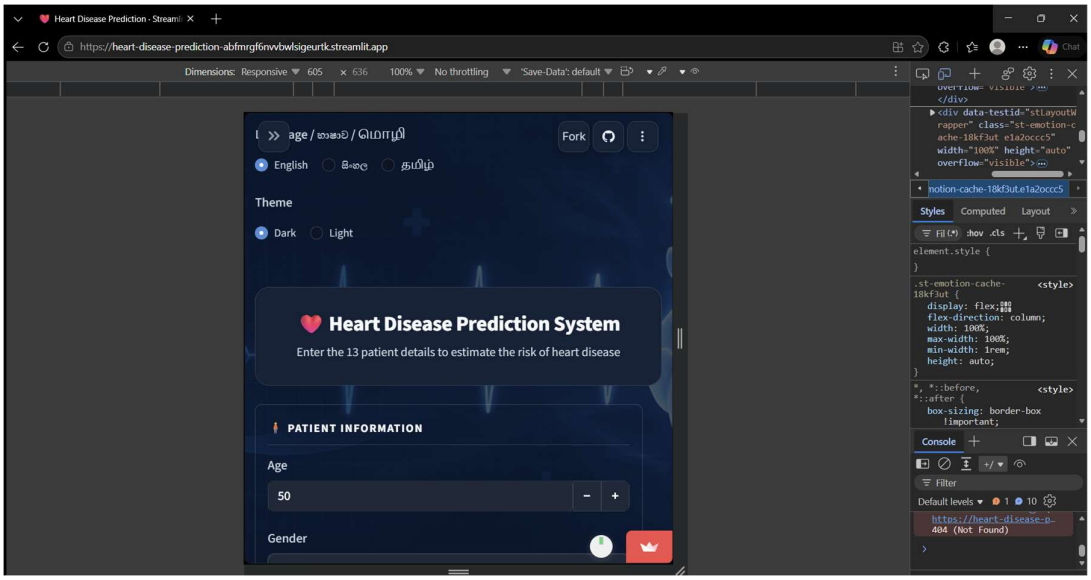


Figure 80: Web app at mobile, tablet and desktop widths

6. System Evaluation and Testing

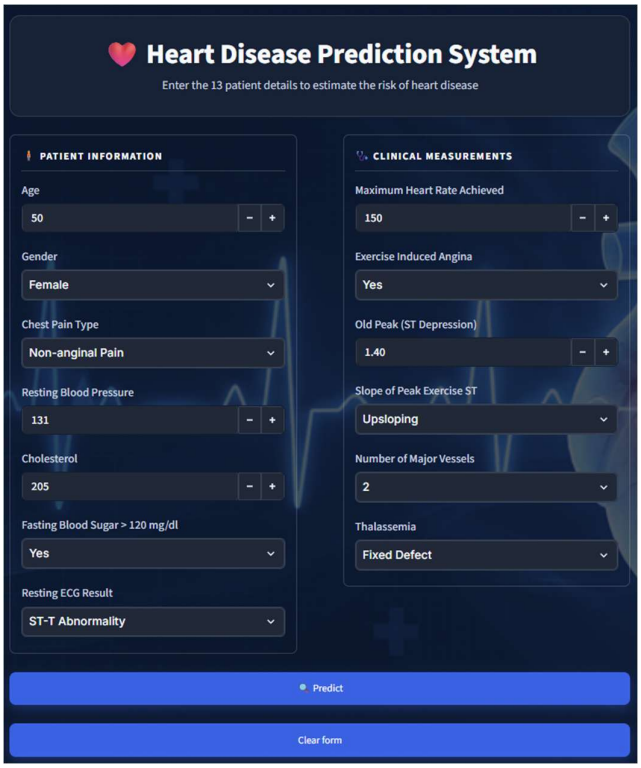


Figure 81: Sample patient input entered in the app



Figure 82: Resulting prediction for the sample patient

7. Repository Maintenance and Merging



Figure 83: GitHub commit history showing merge and conflict-fix commits

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:		**Date:**	17/10/2026

Student Name	RMUT Rathnayake	Registration No.	D/BIT/24/0048
Degree / Intake	BSc (Hons) in Information Technology Intake 41	Group No.	42
Project Title	AI-Based Heart Disease Risk Prediction System		

Technical Contributions

Table 12: Technical contributions – RMUT Rathnayake

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
GUI Testing	Tested the application interface by entering different patient inputs (default, typical, one-variable change, boundary and extreme values) and checked that the prediction, results page, sidebar and PDF report download worked correctly.	Streamlit web interface, Google Chrome; black-box (functional) testing	Test case table with inputs, expected and actual results, confidence and Pass/Fail; input and result screenshots
Interface Enhancements	Added a sidebar with a Dark/Light mode option and sections that explain what the system is about, how to use it and a disclaimer. Added a background image to the interface.	Python, Streamlit, VS Code	Updated interface (app.py); dark and light modes; background image
Result Page	Added a separate result page that opens after Predict. It shows whether heart disease is detected or not, the confidence percentage and the risk level, and lists the patient details used for the prediction.	Python, Streamlit	Updated interface (app.py); screenshots of the result page with the prediction, confidence, risk level and patient details
Testing Methodology and Sample Outputs	Documented the testing methodology, including the baseline patient and test approach, and recorded sample outputs and observations from the test.	Black-box testing	Methodology section, sample output screenshots and recorded test results
Project Proposal, Project Review and Final Report Documentation	Prepared and reviewed sections of the project proposal, project review and final report, covering the expected inputs and outputs, proposed tools and technologies, development problems and system testing documentation.	Collaborative document preparation; technical writing; test case documentation	Project proposal; expected inputs and outputs, tools and technologies, system diagram Final report; testing section with test tables and screenshots

Evidence

1. Interface Enhancements – Sidebar, Dark Mode, Light Mode and Background Image

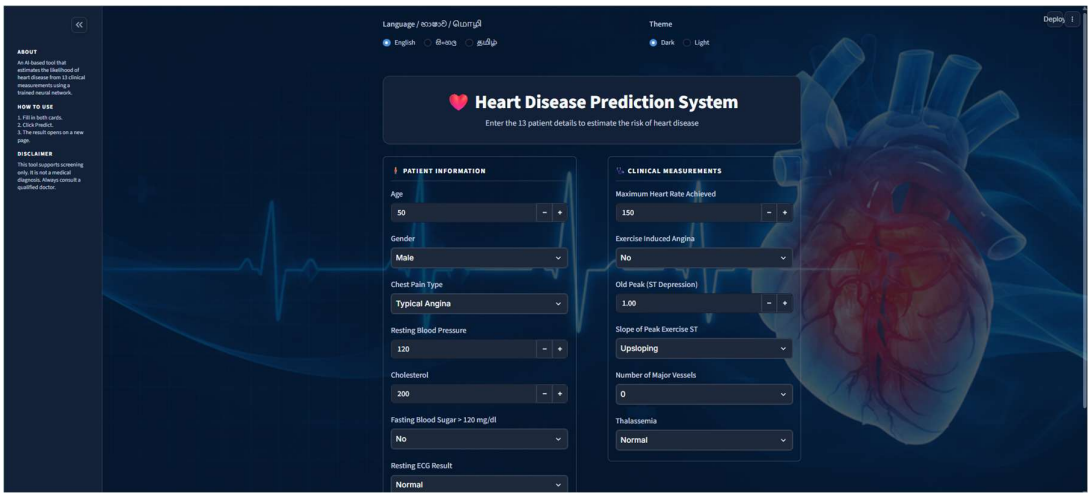


Figure 84: Web app in dark mode with the sidebar and background image

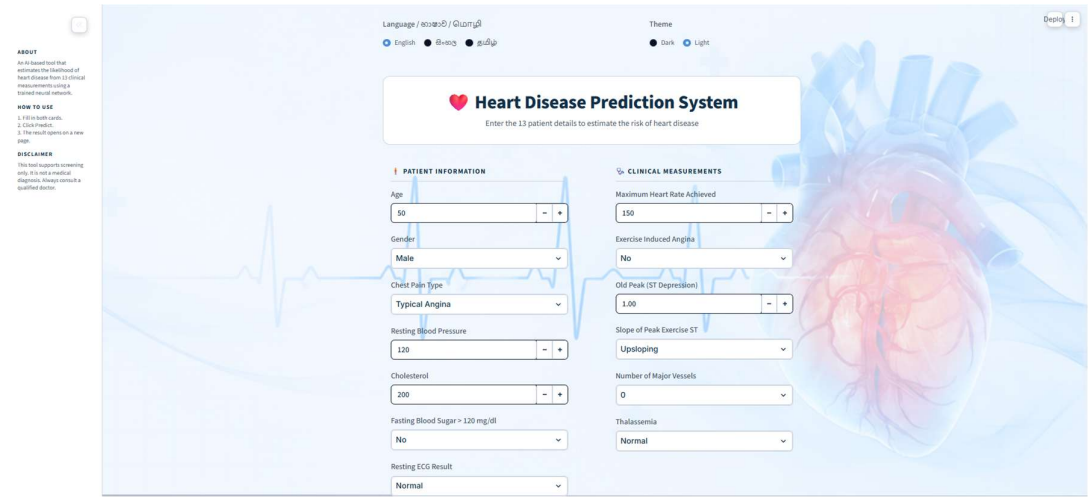


Figure 85: Web app in light mode with the sidebar and background image



Figure 86: Sidebar code (About, How to use and Disclaimer sections) in app.py

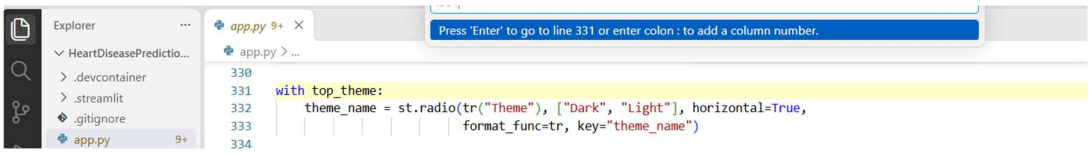


Figure 87: Dark/Light theme selector code in app.py



Figure 88: Background image loading function in app.py



Figure 89: Background image style rule in app.py

2. Result Page – Progress Indicator, Risk Level and Patient Details

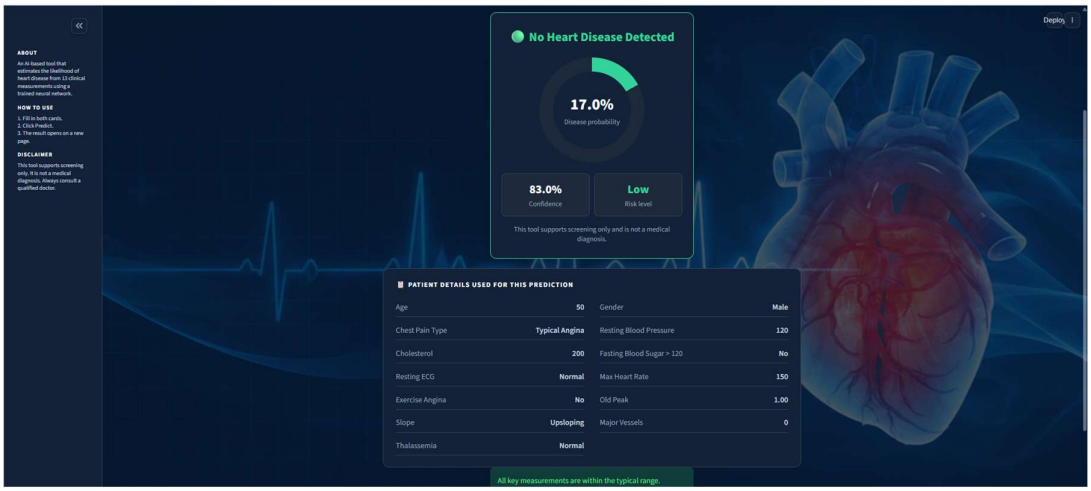
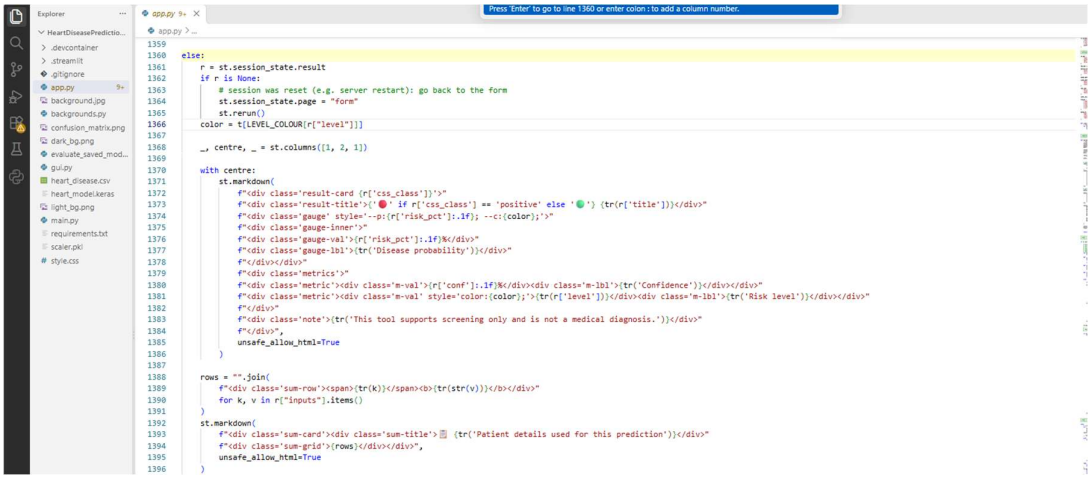


Figure 90: Result page showing the prediction, confidence, risk level and patient details used



```
else:
    r = st.session_state.result
    if r is None:
        # session was reset (e.g. server restart): go back to the form
        st.session_state.page = "form"
        st.rerun()
    color = t[LEVEL_COLOUR[r["level"]]]

    _, centre, _ = st.columns([1, 2, 1])

    with centre:
        st.markdown(
            f"<div class='result-card {r['css_class']}'>"
            f"<div class='result-title'>{'🔴' if r['css_class'] == 'positive' else '🟢'} {tr(r['title'])}</div>"
            f"<div class='gauge' style='--p:{r['risk_pct']:.1f}; --c:{color};'>"
            f"<div class='gauge-inner'>"
            f"<div class='gauge-val'>{r['risk_pct']:.1f}%</div>"
            f"<div class='gauge-lbl'>{tr('Disease probability')}</div>"
            f"</div></div>"
            f"<div class='metrics'>"
            f"<div class='metric'><div class='m-val'>{r['conf']:.1f}%</div><div class='m-lbl'>{tr('Confidence')}</div></div>"
            f"<div class='metric'><div class='m-val' style='color:{color};'>{tr(r['level'])}</div><div class='m-lbl'>{tr('Risk level')}</div></div>"
            f"</div>"
            f"<div class='note'>{tr('This tool supports screening only and is not a medical diagnosis.')}</div>"
            f"</div>",
            unsafe_allow_html=True
        )

    rows = "".join(
        f"<div class='sum-row'><span>{tr(k)}</span><b>{tr(str(v))}</b></div>"
        for k, v in r["inputs"].items()
    )
    st.markdown(
        f"<div class='sum-card'><div class='sum-title'>📋 {tr('Patient details used for this prediction')}</div>"
        f"<div class='sum-grid'>{rows}</div></div>",
        unsafe_allow_html=True
    )
```

Figure 91: Result page code in app.py

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:		**Date:**	17/10/2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.

Student Name	G.I.D de Zoysa	Registration No.	D/BIT/24/0003
Degree / Intake	BSc (Hons) in Information Technology Intake 41	Group No.	42
Project Title	AI-Based Heart Disease Risk Prediction System		

Technical Contributions

Table 13: Technical contributions – G.I.D de Zoysa

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
GUI Layout and Structure	Designed and organized the interface into clear patient information, clinical input and prediction sections, with intuitive controls.	Python; Streamlit; Tkinter/ttk	Application screens; app.py; gui.py
Patient Input Forms	Implemented the 13 patient/clinical feature inputs, dropdowns and prediction/reset controls to support data entry.	Python; Streamlit; Tkinter/ttk	Patient input form and controls
Input Validation and User Guidance	Worked on required-field checks, numeric input handling, error messages and help tooltips to guide users through data entry.	Python; Streamlit	Form validation and help text
Interface Styling and Options	Added responsive interface styling, Light/Dark themes and language options to improve usability and readability.	Streamlit; HTML/CSS	Theme, language and layout screens
Model Integration and Prediction Flow	Connected GUI inputs to the trained prediction model and saved scaler, with feature preparation for model inference.	Python; TensorFlow/Keras; scikit-learn; joblib	Prediction workflow in app.py and gui.py
Prediction Results and PDF Output	Implemented the result presentation, probability gauge, what-if analysis, key factors, session history and downloadable PDF report.	Python; Streamlit; pandas; NumPy; ReportLab	Prediction result screens; generated PDF
Project Documentation (Proposal, Progress Review and Final Report)	Contributed to the three project documents, covering GUI requirements, design decisions, development progress, interface features, prediction integration and testing.	Technical writing; MS Word; team review	Project proposal; progress review report; final report
GUI Manual Testing and Validation	Prepared and carried out 22 manual GUI test cases for input handling, navigation, language/theme options, predictions, report download, responsiveness and end-to-end use. Recorded the self-reported test outcomes.	Manual functional testing; Excel / Google Sheets	GUI testing log; 22 tester-reported Pass results

Evidence

1. GUI Design and Implementation

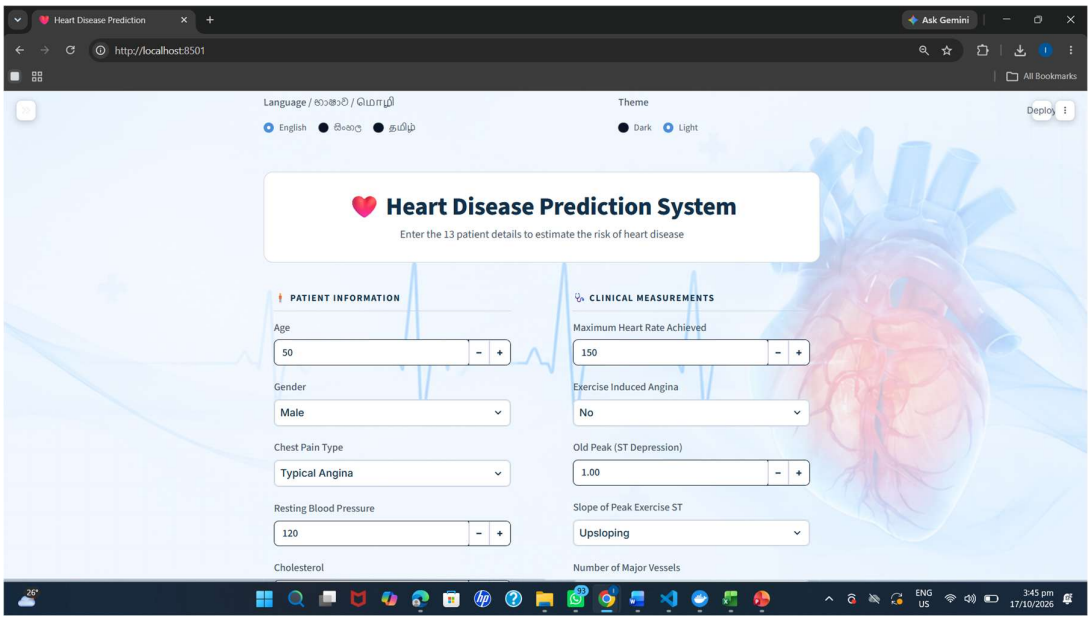


Figure 92: Patient information and clinical measurements interface



Figure 93: Prediction result and patient details

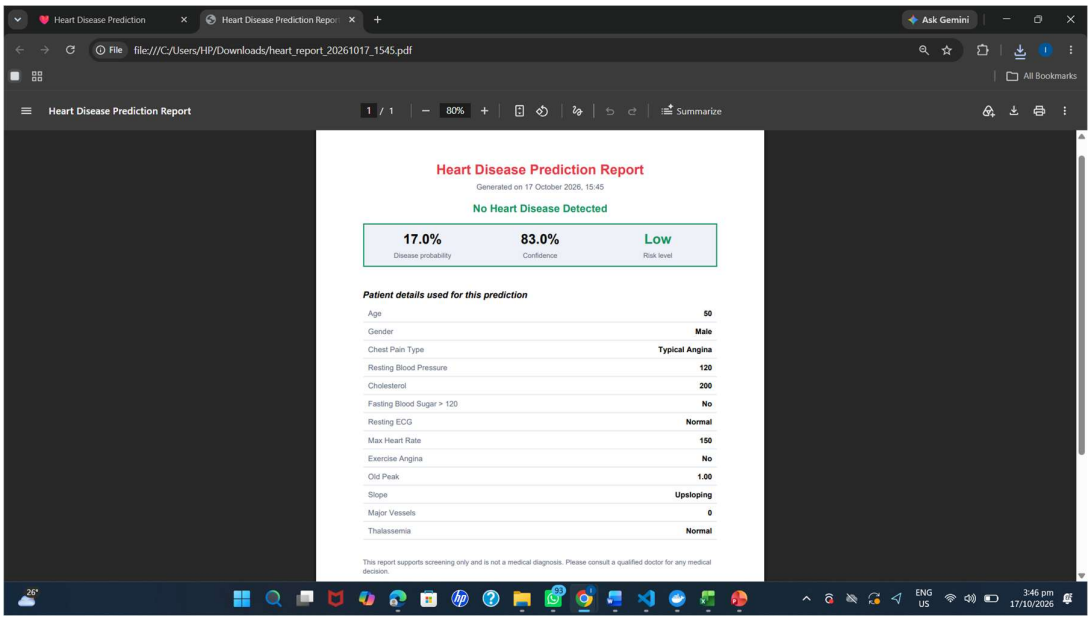


Figure 94: Generated PDF prediction report

2. GUI Source Code and Integration

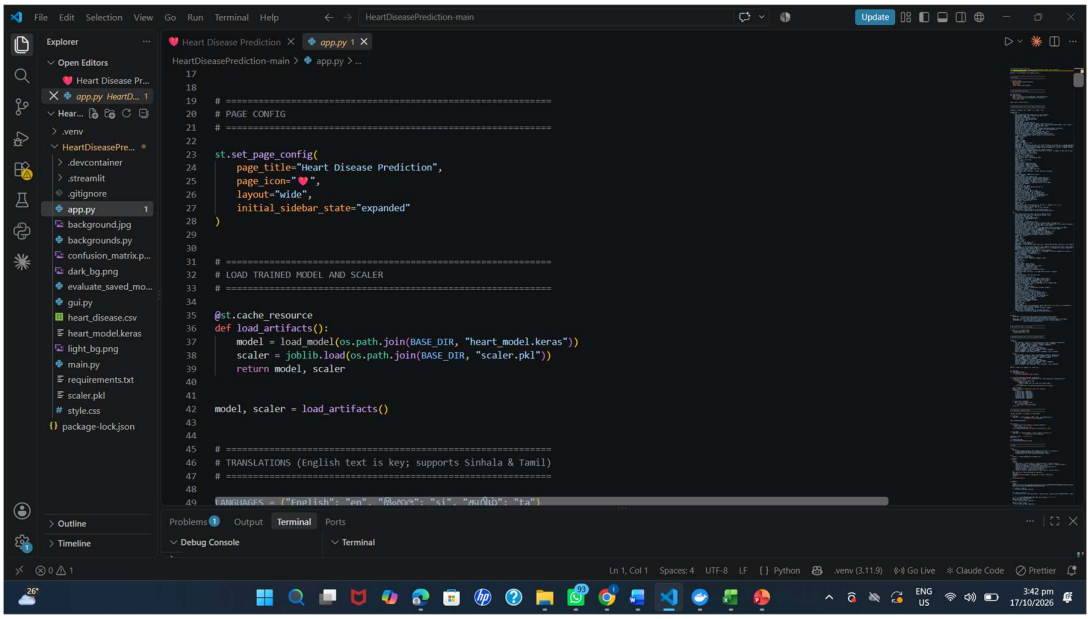


Figure 95: Streamlit page configuration and model loading

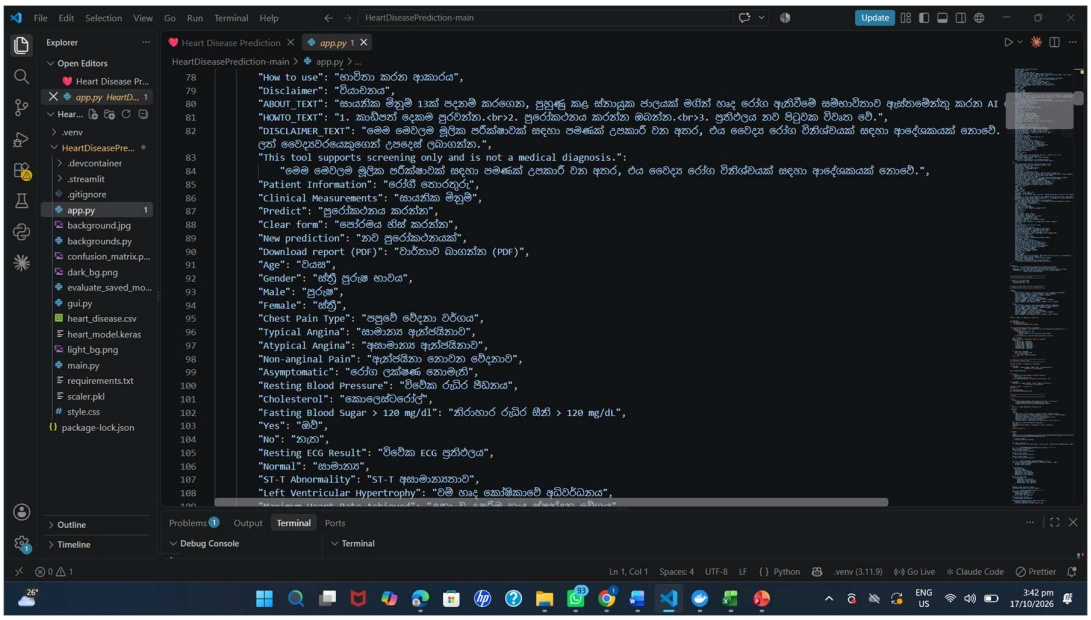


Figure 96: Sinhala translation definitions in the application

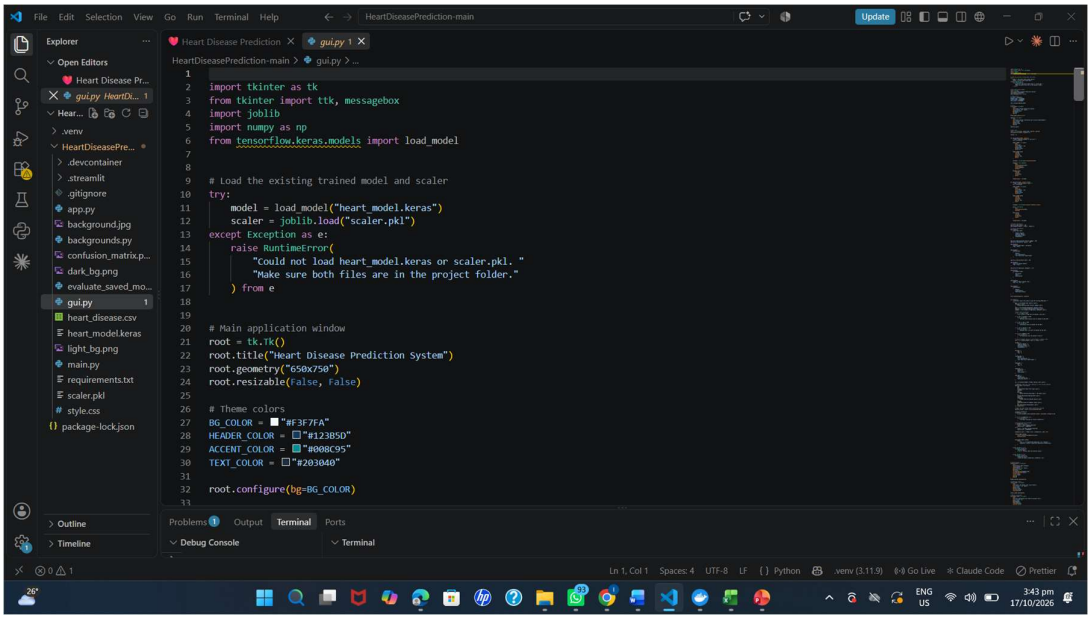
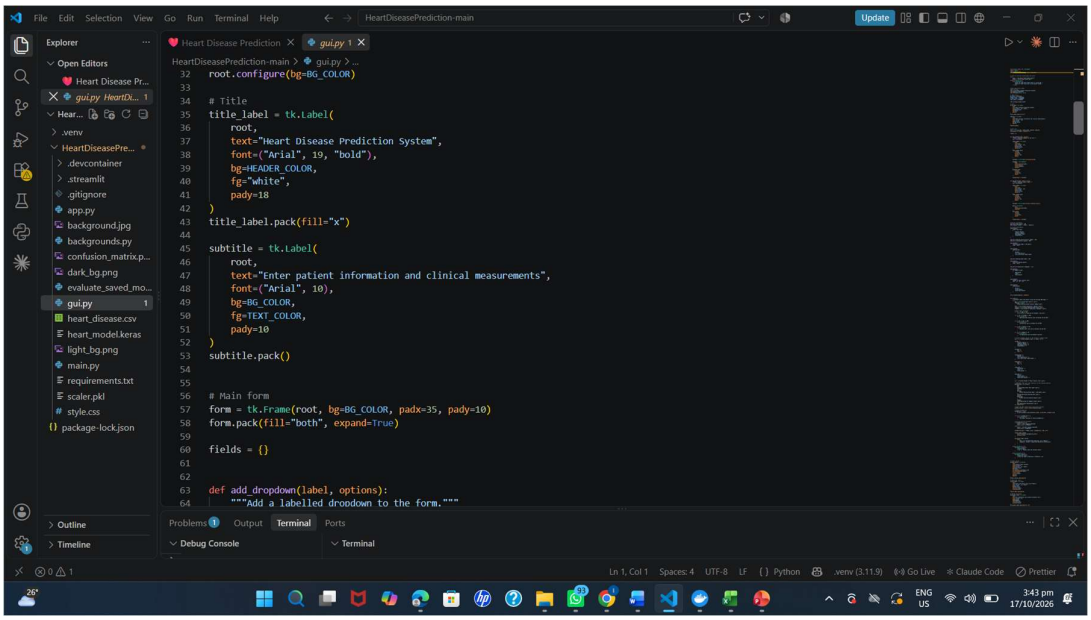


Figure 97: Tkinter GUI setup and model loading



```
root.configure(bg=BG_COLOR)

# Title
title_label = tk.Label(
    root,
    text="Heart Disease Prediction System",
    font=("Arial", 19, "bold"),
    bg=HEADER_COLOR,
    fg="white",
    pady=18
)
title_label.pack(fill="x")

subtitle = tk.Label(
    root,
    text="Enter patient information and clinical measurements",
    font=("Arial", 10),
    bg=BG_COLOR,
    fg=TEXT_COLOR,
    pady=10
)
subtitle.pack()

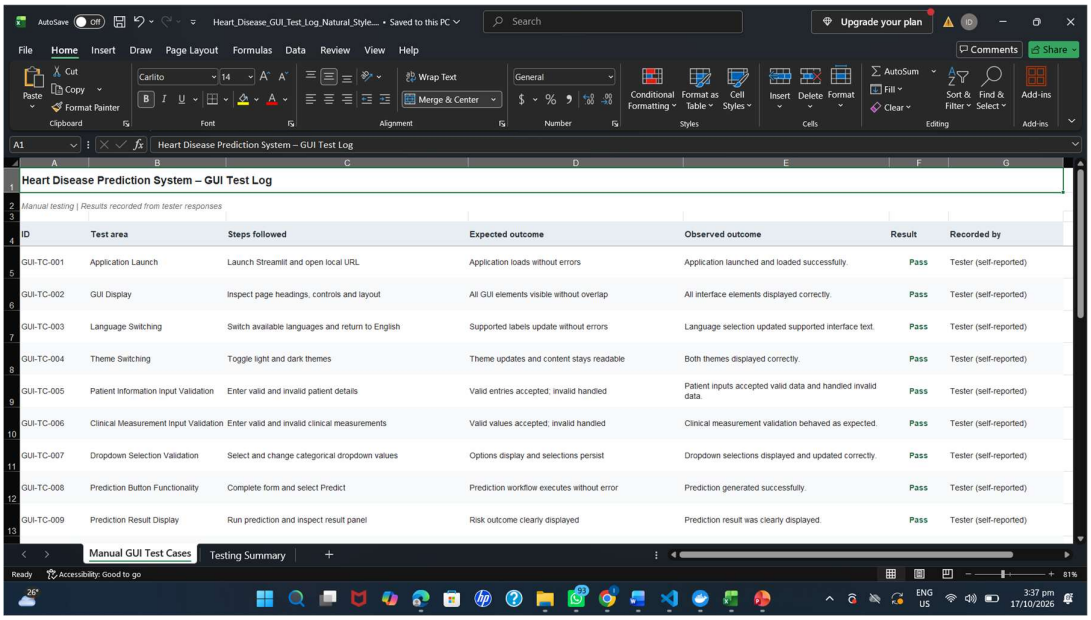

# Main form
form = tk.Frame(root, bg=BG_COLOR, padx=35, pady=10)
form.pack(fill="both", expand=True)

fields = {}


def add_dropdown(label, options):
    """Add a labelled dropdown to the form."""
```

Figure 98: Tkinter interface layout and form controls

3. GUI Testing and Validation

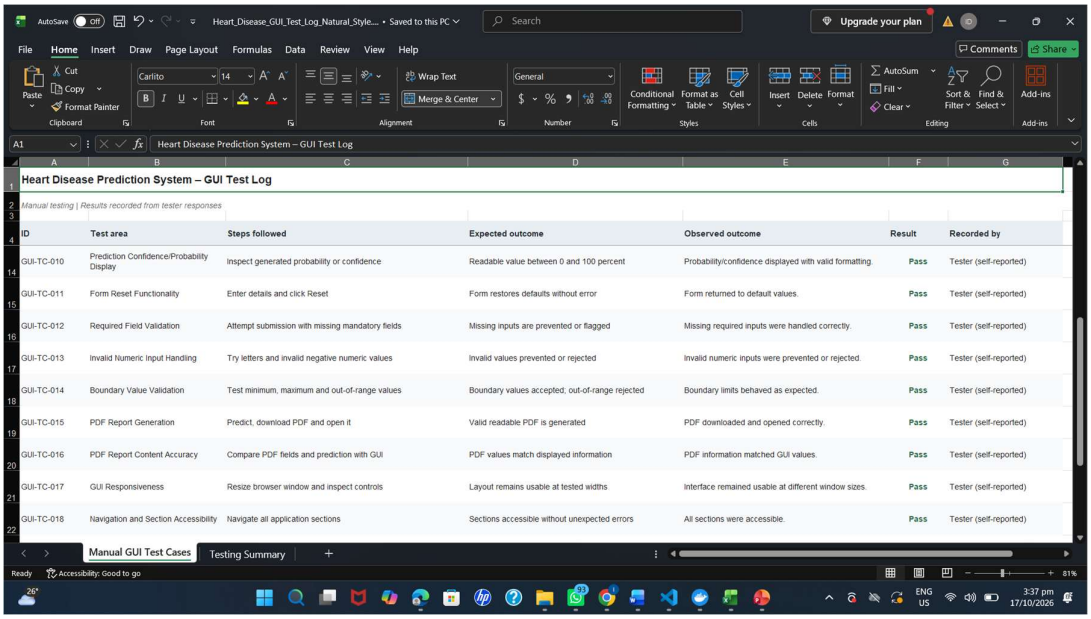


Heart Disease Prediction System – GUI Test Log

Manual testing | Results recorded from tester responses

ID	Test area	Steps followed	Expected outcome	Observed outcome	Result	Recorded by
GUI-TC-001	Application Launch	Launch Streamlit and open local URL	Application loads without errors	Application launched and loaded successfully.	Pass	Tester (self-reported)
GUI-TC-002	GUI Display	Inspect page headings, controls and layout	All GUI elements visible without overlap	All interface elements displayed correctly.	Pass	Tester (self-reported)
GUI-TC-003	Language Switching	Switch available languages and return to English	Supported labels update without errors	Language selection updated supported interface text.	Pass	Tester (self-reported)
GUI-TC-004	Theme Switching	Toggle light and dark themes	Theme updates and content stays readable	Both themes displayed correctly.	Pass	Tester (self-reported)
GUI-TC-005	Patient Information Input Validation	Enter valid and invalid patient details	Valid entries accepted; invalid handled	Patient inputs accepted valid data and handled invalid data.	Pass	Tester (self-reported)
GUI-TC-006	Clinical Measurement Input Validation	Enter valid and invalid clinical measurements	Valid values accepted; invalid handled	Clinical measurement validation behaved as expected.	Pass	Tester (self-reported)
GUI-TC-007	Dropdown Selection Validation	Select and change categorical dropdown values	Options display and selections persist	Dropdown selections displayed and updated correctly.	Pass	Tester (self-reported)
GUI-TC-008	Prediction Button Functionality	Complete form and select Predict	Prediction workflow executes without error	Prediction generated successfully.	Pass	Tester (self-reported)
GUI-TC-009	Prediction Result Display	Run prediction and inspect result panel	Risk outcome clearly displayed	Prediction result was clearly displayed.	Pass	Tester (self-reported)

Figure 99: GUI test log: test cases 01–09

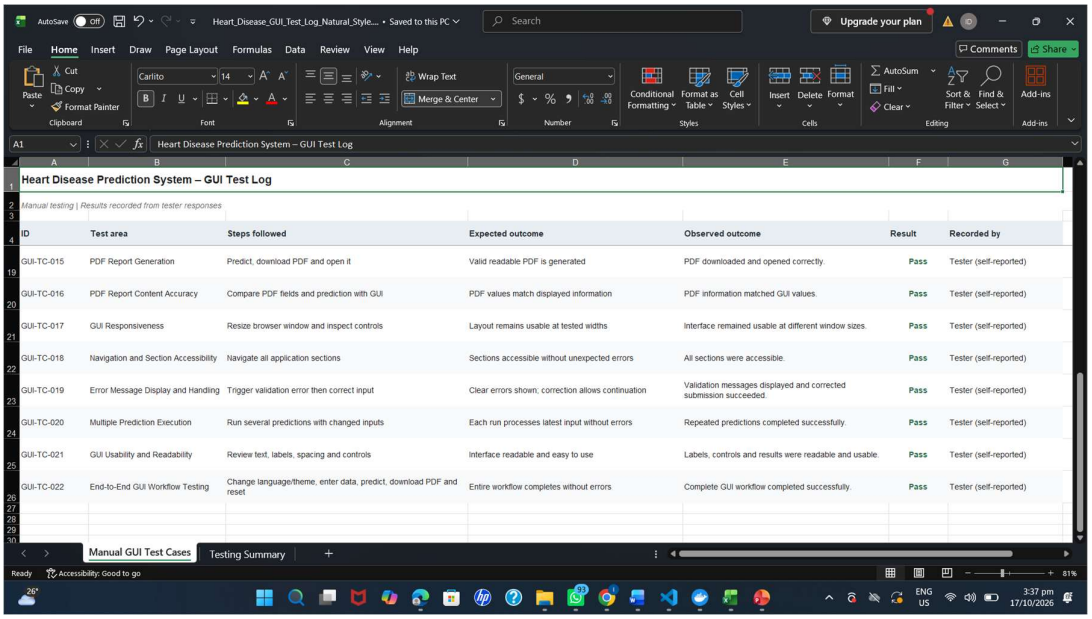


Heart Disease Prediction System – GUI Test Log

Manual testing | Results recorded from tester responses

ID	Test area	Steps followed	Expected outcome	Observed outcome	Result	Recorded by
GUI-TC-010	Prediction Confidence/Probability Display	Inspect generated probability or confidence	Readable value between 0 and 100 percent	Probability/confidence displayed with valid formatting.	Pass	Tester (self-reported)
GUI-TC-011	Form Reset Functionality	Enter details and click Reset	Form restores defaults without error	Form returned to default values.	Pass	Tester (self-reported)
GUI-TC-012	Required Field Validation	Attempt submission with missing mandatory fields	Missing inputs are prevented or flagged	Missing required inputs were handled correctly.	Pass	Tester (self-reported)
GUI-TC-013	Invalid Numeric Input Handling	Try letters and invalid negative numeric values	Invalid values prevented or rejected	Invalid numeric inputs were prevented or rejected.	Pass	Tester (self-reported)
GUI-TC-014	Boundary Value Validation	Test minimum, maximum and out-of-range values	Boundary values accepted; out-of-range rejected	Boundary limits behaved as expected.	Pass	Tester (self-reported)
GUI-TC-015	PDF Report Generation	Predict, download PDF and open it	Valid readable PDF is generated	PDF downloaded and opened correctly.	Pass	Tester (self-reported)
GUI-TC-016	PDF Report Content Accuracy	Compare PDF fields and prediction with GUI	PDF values match displayed information	PDF information matched GUI values.	Pass	Tester (self-reported)
GUI-TC-017	GUI Responsiveness	Resize browser window and inspect controls	Layout remains usable at tested widths	Interface remained usable at different window sizes.	Pass	Tester (self-reported)
GUI-TC-018	Navigation and Section Accessibility	Navigate all application sections	Sections accessible without unexpected errors	All sections were accessible.	Pass	Tester (self-reported)

Figure 100: GUI test log: test cases 10–18

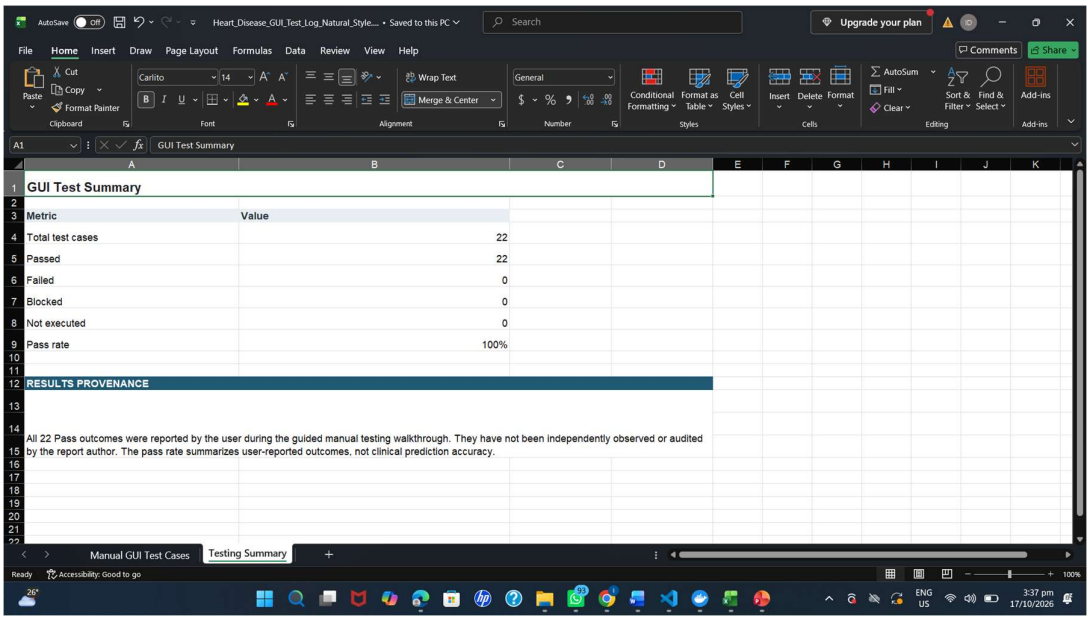


Heart Disease Prediction System – GUI Test Log

Manual testing | Results recorded from tester responses

ID	Test area	Steps followed	Expected outcome	Observed outcome	Result	Recorded by
GUI-TC-015	PDF Report Generation	Predict, download PDF and open it	Valid readable PDF is generated	PDF downloaded and opened correctly.	Pass	Tester (self-reported)
GUI-TC-016	PDF Report Content Accuracy	Compare PDF fields and prediction with GUI	PDF values match displayed information	PDF information matched GUI values.	Pass	Tester (self-reported)
GUI-TC-017	GUI Responsiveness	Resize browser window and inspect controls	Layout remains usable at tested widths	Interface remained usable at different window sizes.	Pass	Tester (self-reported)
GUI-TC-018	Navigation and Section Accessibility	Navigate all application sections	Sections accessible without unexpected errors	All sections were accessible.	Pass	Tester (self-reported)
GUI-TC-019	Error Message Display and Handling	Trigger validation error then correct input	Clear errors shown; correction allows continuation	Validation messages displayed and corrected submission succeeded.	Pass	Tester (self-reported)
GUI-TC-020	Multiple Prediction Execution	Run several predictions with changed inputs	Each run processes latest input without errors	Repeated predictions completed successfully.	Pass	Tester (self-reported)
GUI-TC-021	GUI Usability and Readability	Review text, labels, spacing and controls	Interface readable and easy to use	Labels, controls and results were readable and usable.	Pass	Tester (self-reported)
GUI-TC-022	End-to-End GUI Workflow Testing	Change language/theme, enter data, predict, download PDF and reset	Entire workflow completes without errors	Complete GUI workflow completed successfully.	Pass	Tester (self-reported)

Figure 101: GUI test log: test cases 15–22



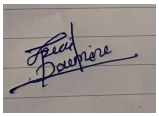
GUI Test Summary		
Metric	Value	
Total test cases	22	
Passed	22	
Failed	0	
Blocked	0	
Not executed	0	
Pass rate	100%	
RESULTS PROVENANCE		
All 22 Pass outcomes were reported by the user during the guided manual testing walkthrough. They have not been independently observed or audited by the report author. The pass rate summarizes user-reported outcomes, not clinical prediction accuracy.		

Figure 102: GUI testing summary: 22 tester-reported passes

Testing note: The 22 Pass results were reported by the tester during guided manual testing; they were not independently audited.

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:		**Date:**	17/10/2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.